Sharp convergence regions and periodic obstructions for Chambolle–Pock

Abstract

For the literal primal-first Chambolle–Pock iteration, we prove the sharp universal strict product cap $R_3(\theta) = (\sqrt{4 + 12\theta + 12\theta^2} + 1 + 3\theta)/(1 + \theta)^2$ on two explicit closed extrapolation intervals, one containing $1/8$ and one containing $1/7$, with a separate exact result at $11/74$. For fixed proper lower-semicontinuous convex objectives on arbitrary real Hilbert spaces, bounded linear coupling, nonempty KKT set and arbitrary initialization, every strict product below this curve gives bounded whole-sequence weak convergence, square-summable increments and strongly vanishing KKT residuals. Convergence is in norm in finite dimensions. Exact rational storage families and continuous Bernstein–Gershgorin bounds certify the two regions; contraction embedding covers every smaller product. The complete original $1/8$ proof and its coefficientwise recovery are retained. For all $\theta \geq 0$, the same curve is the exact minimum product for an actual nonconstant three-periodic orbit, with an explicit fixed-triangle realization. Nevertheless, two fixed indicator functions in $\mathbb{R}^4$, with a unique KKT pair, admit a least-period-five orbit at $\theta = 9/50$, product $147/50$, strictly below both this curve and the bilinear period-two obstruction. The same states and hulls work on an explicit parameter rectangle. Thus period-two and period-three obstructions alone do not determine the full convergence boundary. The claims are supported by exact data and independent solver-free replays; neither a complete extrapolation classification nor endpoint convergence is asserted. On $[177/1000, 183/1000]$, we further identify the exact five-cycle onset as an analytic algebraic branch. Two unique transverse crossings delimit precisely where it preempts periods two through four, and every normalized endpoint moment vector is the same saturated two-frequency vector.

Keywords: primal–dual splitting; Chambolle–Pock; weak convergence; exact Lyapunov certificate; sharp step size; convex interpolation; periodic orbit

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Guide to the results

Theorem 1 and Sections 1–6 retain the complete fixed $\theta = 1/8$ result and proof. In particular, (2) still denotes that fixed-parameter algorithm and Theorem 1 remains Theorem 1. Their scope statements concern the original certificate. The general-parameter algorithm is (51); Theorem 2 gives its original sharp interval result. Theorems 3 and 4 establish the all-extrapolation period-three construction and exclusion. Section 10 extends the sufficient interval without changing its certificate, adds a different certificate near $1/7$, and constructs an actual fixed-convex period-five obstruction beyond these sufficient regions. The original theorem is a valid special case, and no proof or historical label has been removed. Appendix A gives the original exact witness; Appendices B and C give

the continuous certificates and additional exact checks. No general convergence assertion is made at $\theta = 0$, at the strict product endpoint, or in an uncertified gap between the stated intervals.

Section 11 classifies this five-cycle onset throughout $[177/1000, 183/1000]$, with the central statement in Theorem 8. Its continuous branch, two exact transverse crossings and endpoint moment rigidity deepen the earlier example. Appendix D describes the portable exact replay. The conic criterion, realization and active-face methods have the antecedents identified at the start of that section.

1 The iteration and the sharp threshold

Let $\mathcal{H}$ and $\mathcal{G}$ be real Hilbert spaces. Let $f : \mathcal{H} \to (-\infty, +\infty]$ and $h : \mathcal{G} \to (-\infty, +\infty]$ be proper lower-semicontinuous convex functions, and let $L : \mathcal{H} \to \mathcal{G}$ be bounded and linear. We use the KKT set

$$\mathcal{Z} = \{(x, y) : -L^*y \in \partial f(x), \quad Lx \in \partial h(y)\}. \tag{1}$$

Thus the usual notation $h = g^*$ is included; conversely, any such h can be written g^* by taking $g = h^*$. For $\tau, \sigma > 0$, the precise algorithm studied here is

$$\begin{aligned} x_{n+1} &= \operatorname{prox}_{\tau f}(x_n - \tau L^* y_n), \\ y_{n+1} &= \operatorname{prox}_{\sigma h}\left(y_n + \sigma L \frac{9x_{n+1} - x_n}{8}\right), \qquad n \geq 0. \end{aligned} \tag{2}$$

There is no outer relaxation, correction step, or varying step size. The proximal mapping convention is $\operatorname{prox}_{\eta f}(a) = \arg\min_x\{\eta f(x) + \|x - a\|^2/2\}$. All these mappings are everywhere defined and single valued.

Set

$$r_\star = \frac{88 + 16\sqrt{91}}{81} = 2.970744101564337\ldots, \qquad 81r_\star^2 - 176r_\star - 192 = 0. \tag{3}$$

Theorem 1. *Suppose $\mathcal{Z} \neq \varnothing$, and initialize (2) at an arbitrary $(x_0, y_0) \in \mathcal{H} \times \mathcal{G}$. If*

$$\rho := \tau\sigma\|L\|^2 < r_\star, \tag{4}$$

then the sequence is bounded and converges weakly to a point of $\mathcal{Z}$. Moreover,

$$\sum_{n=0}^{\infty} \left(\|x_{n+1} - x_n\|^2 + \|y_{n+1} - y_n\|^2\right) < \infty. \tag{5}$$

For $n \geq 1$, the residual pair

$$\left(-\frac{x_n - x_{n-1}}{\tau} + L^*(y_n - y_{n-1}),\ -\frac{y_n - y_{n-1}}{\sigma} + \frac{1}{8}L(x_n - x_{n-1})\right) \tag{6}$$

belongs to $(\partial f(x_n) + L^* y_n) \times (\partial h(y_n) - L x_n)$ *and tends to zero in norm. If both spaces are finite dimensional, the iterates converge in norm.*

The constant $r_\star$ *is sharp as a universal strict upper bound: there is an instance in a real Hilbert plane with nonempty KKT set and* $\tau\sigma\|L\|^2 = r_\star$ *for which* (2) *has a nonconstant period-three orbit.*

The universal quantifiers in this statement matter. An upper bound larger than $r_\star$, used with a strict inequality, would include the endpoint counterexample and is therefore impossible. The theorem does not say that every instance at or above $r_\star$ fails. In particular, it gives no universal boundedness conclusion at the endpoint, and its proof does not give infinite-dimensional strong convergence.

1.1 Relation to existing results

The Chambolle–Pock primal–dual method originates in [1]. The closest explicit small-extrapolation theorem for the same literal map and Hilbert-space assumptions that we compare here is [2, Theorem 1.2]. Its sufficient product bound for $0 < \theta \le 1/2$ is

$$\tau\sigma\|L\|^2 < \frac{4\theta(2-\theta)}{1 - 2\theta + 9\theta^2 - 4\theta^3}. \tag{7}$$

At $\theta = 1/8$ this is $120/113 = 1.0619469026\ldots$. The sharp sufficient bound $4/(1+2\theta)$ in [3, Theorem 2] is proved for $\theta > 1/2$; it cannot be extended to $\theta = 1/8$ by substitution. The scalar bilinear example discussed in that work does yield the necessary ceiling $16/5$ at $\theta = 1/8$. It is strictly larger than $r_\star$; the endpoint obstruction here is nonlinear.

Longer-memory convex-interpolation Lyapunov constructions are prior methodology. In particular, [4, Section 3.1 and Figure 2] explores numerical SDP parameter regions for this map with several memory and overlap choices, and reports squared fixed-point-residual summability certificates. The present argument uses an exact algebraic family, a whole-interval positivity proof, a general-coupling Hilbert-space convergence argument, and a matching endpoint cycle. We do not claim that the longer-memory method or numerical evidence for larger small-extrapolation steps originates here. The comparison is to the specified sources and versions, not an exhaustive publication-priority claim. The full proof below does not depend on any numerical SDP status or any sampled eigenvalue.

2 Normalization and actual convexity inequalities

2.1 One fixed parameter for the whole orbit

Let $\delta = 1/100000$. Under (4), choose once and for all

$$t = \min\left\{\delta, \frac{r_\star - \rho}{2}\right\}, \qquad r = r_\star - t. \tag{8}$$

Then $0 < t \le \delta$, $r > \rho$, and $r > 0$. This parameter is fixed throughout the proof; neither the algorithm nor the certificate changes with n. Define

$$X = \sqrt{r/\tau}\, x, \quad Y = y/\sqrt{\sigma}, \quad K = \sqrt{\tau\sigma/r}\, L, \tag{9}$$

and

$$F(X) = f(\sqrt{\tau/r}\, X), \qquad H(Y) = h(\sqrt{\sigma}\, Y).$$

Here $K : \mathcal{H} \to \mathcal{G}$ is a contraction. The normalized iterates satisfy, for $n \ge 1$,

$$\begin{aligned} (X_{n-1} - X_n)/r - K^* Y_{n-1} &\in \partial F(X_n), \\ Y_{n-1} - Y_n + K(9X_n - X_{n-1})/8 &\in \partial H(Y_n). \end{aligned} \tag{10}$$

These follow directly from the two original optimality inclusions and the subdifferential rule for a positive scalar change of coordinates. For example, multiplication of the primal inclusion by $\sqrt{\tau/r}$ gives exactly the first line. The normalized KKT pairs satisfy $-K^*V \in \partial F(U)$ and $KU \in \partial H(V)$. They correspond bijectively to (1) under (9).

This reduction also covers $L = 0$. There is no inverse of L, no lower bound on its norm, and no dimension assumption. The role of the normalization is important: a certificate on just the interval $r_\star - \delta \le r < r_\star$ will imply the theorem for every actual product $\rho < r_\star$.

2.2 Anchored values and subgradients

Fix any normalized KKT pair $p = (U, V)$ and write $u_n = X_n - U$, $v_n = Y_n - V$. Define shifted functions

$$\begin{aligned} F_p(u) &= F(U + u) - F(U) + \langle K^*V, u\rangle, \\ H_p(v) &= H(V + v) - H(V) - \langle KU, v\rangle. \end{aligned} \tag{11}$$

They are nonnegative, vanish at zero, and have zero as a subgradient there. The actual subgradients at positive-index iterates include

$$\begin{aligned} a_n &= (u_{n-1} - u_n)/r - K^* v_{n-1} \in \partial F_p(u_n), \\ b_n &= v_{n-1} - v_n + K(9u_n - u_{n-1})/8 \in \partial H_p(v_n). \end{aligned} \tag{12}$$

Every positive-index function value is finite, because X_n, Y_n are proximal outputs. We make no such assumption at $n = 0$.

Convexity therefore gives, for positive i, j,

$$\begin{aligned} F_p(u_j) - F_p(u_i) - \langle a_i, u_j - u_i\rangle &\ge 0, \\ H_p(v_j) - H_p(v_i) - \langle b_i, v_j - v_i\rangle &\ge 0. \end{aligned} \tag{13}$$

The anchor gives the additional inequalities

$$\begin{aligned} 0 \le F_p(u_i) &\le \langle a_i, u_i\rangle, \\ 0 \le H_p(v_i) &\le \langle b_i, v_i\rangle. \end{aligned} \tag{14}$$

These are all genuine subgradient inequalities of the original functions after normalization and anchoring. No independent interpolation hypothesis is imposed on the iterates.

2.3 Embedding the scalar certificate for a contraction

All formal certificate slots lie in one Hilbert space. To transfer the certificate to the contraction K, use $\mathcal{E} = \mathcal{G} \oplus \mathcal{H}$ and isometries

$$Ju = (Ku, (I - K^*K)^{1/2}u), \qquad Iv = (v, 0). \tag{15}$$

The positive square root exists since $K^*K \le I$, and

$$\langle Ju, Ju' \rangle = \langle u, u' \rangle, \quad \langle Iv, Iv' \rangle = \langle v, v' \rangle, \quad \langle Ju, Iv \rangle = \langle Ku, v \rangle. \tag{16}$$

For a five-pair segment starting at $k \ge 1$, substitute

$$\boldsymbol{X}_i = Ju_{k+i}, \quad \boldsymbol{Y}_i = Iv_{k+i}, \quad \phi_{i,0} = F_p(u_{k+i}), \quad \phi_{i,1} = H_p(v_{k+i}), \qquad 0 \le i \le 4. \tag{17}$$

In formal slots the gradient expressions are

$$\boldsymbol{a}_i = (\boldsymbol{X}_{i-1} - \boldsymbol{X}_i)/r - \boldsymbol{Y}_{i-1}, \qquad \boldsymbol{b}_i = \boldsymbol{Y}_{i-1} - \boldsymbol{Y}_i + (9\boldsymbol{X}_i - \boldsymbol{X}_{i-1})/8. \tag{18}$$

Every pairing of a formal gradient with a same-variable point difference is, by (16), exactly the corresponding pairing in (13). Hence every formal slack used below is nonnegative on (17). This argument preserves the needed pairings; it does not assert that the formal gradients are subgradients of invented functions on $\mathcal{E}$.

3 An exact four-pair certificate

We state the finite algebraic certificate and prove how its signs are certified. Its coefficient data, reconstruction conventions, and checking algorithms are completely specified in Appendix A. All field operations in that appendix are exact.

For a real symmetric matrix B and Hilbert-valued slots w_i, we use

$$\langle w, Bw \rangle = \sum_{i,j} B_{ij} \langle w_i, w_j \rangle.$$

Let $w = (\boldsymbol{X}_0, \boldsymbol{Y}_0, \dots, \boldsymbol{X}_4, \boldsymbol{Y}_4)$, and let ϕ be the corresponding ten-value vector. Let O select the first eight slots and N the last eight slots. Each formal convexity slack has the form

$$S_e = \langle w, Q_e(r)w \rangle + v_e^T \phi. \tag{19}$$

The set $\mathcal{A}$ has 50 edges; its 32-edge subset $\mathcal{B}$ excludes all edges touching index 4. Both are specified in Appendix A.2.

Lemma 1 (Exact certificate)**.** *On $0 \le t \le \delta$ and $r = r_\star - t$, there are a symmetric 8×8 matrix $P(t)$, a vector $c(t) \in \mathbb{R}^8$, and nonnegative multiplier vectors $\lambda(t) \in \mathbb{R}^{50}$, $\mu(t) \in \mathbb{R}^{32}$*

with entries in $\mathbb{Q}(\sqrt{91})(t)$ *such that*

$$\begin{aligned} D(t) &= O^T P(t) O - N^T P(t) N - \sum_{e\in\mathcal{A}} \lambda_e(t) Q_e(r), \\ C(t) &= O^T P(t) O - \sum_{e\in\mathcal{B}} \mu_e(t) Q_e(r), \end{aligned} \tag{20}$$

while

$$O^T c - N^T c - \sum_{e\in\mathcal{A}} \lambda_e v_e = 0, \qquad O^T c - \sum_{e\in\mathcal{B}} \mu_e v_e = 0. \tag{21}$$

The last two rows and columns of C *vanish. Put*

$$Z = [I_2; I_2; I_2; I_2; I_2], \qquad Z_8 = [I_2; I_2; I_2; I_2].$$

Then $D(t)Z = 0$. *For every* $0 < t \le \delta$, *the leading* 8×8 *blocks of* $rD(t)$ *and* $rC(t)$ *are positive definite.*

If C_8 *denotes the leading block of* C *and* $M = Z_8^T P Z_8$, *then*

$$M_{11} = (Z_8^T C_8 Z_8)_{11}, \qquad M_{22} = (Z_8^T C_8 Z_8)_{22}. \tag{22}$$

At $t = 0$ *one has* $M = 0$.

Proof. The supplied coefficient data take the form

$$P(t) = \frac{P_0 + tP_1 + t^2 P_2}{r_\star - t}, \quad c(t) = c_0 + tc_1, \quad \lambda(t) = \lambda_0 + t\lambda_1, \quad \mu(t) = \mu_0 + t\mu_1. \tag{23}$$

All coefficient entries belong to $\mathbb{Q}(\sqrt{91})$. Reconstruction of the slacks from (18) gives the two quadratic polynomial matrices rD, rC. The two value identities, all zero rows, the constant kernel, and (22) are checked coefficient by coefficient. Nonnegativity of every multiplier is checked at $t = 0$ and $t = \delta$; affinity proves it everywhere in between. Positivity of r follows from $r_\star - \delta > 0$.

Here is the complete interval argument for each leading polynomial block $Q(t) = Q_0 + tQ_1 + t^2 Q_2$. The explicitly specified invertible basis $T = [E, K_0]$ has $Q_0 K_0 = 0$ and gives

$$T^T Q(t) T = \begin{pmatrix} A_0 + tA_1 + t^2 A_2 & t(B_1 + tB_2) \\ t(B_1 + tB_2)^T & t(C_1 + tC_2) \end{pmatrix}. \tag{24}$$

Define

$$H(t) = \begin{pmatrix} A_0 + tA_1 + t^2 A_2 & \delta(B_1 + tB_2) \\ \delta(B_1 + tB_2)^T & \delta(C_1 + tC_2) \end{pmatrix} = H_0 + tH_1 + t^2 H_2. \tag{25}$$

The three exact matrices

$$B^{[0]} = H_0, \qquad B^{[1]} = H_0 + \frac{\delta}{2} H_1, \qquad B^{[2]} = H_0 + \delta H_1 + \delta^2 H_2 \tag{26}$$

have positive exact LDL^T pivots. In total there are six 8×8 positive-definiteness checks, two families of three. The coefficient data and both verifiers reproduce every pivot using rational arithmetic in $\mathbb{Q}(\sqrt{91})$.

With $s = t/\delta \in [0,1]$, one has the polynomial identity

$$H(t) = (1-s)^2 B^{[0]} + 2s(1-s)B^{[1]} + s^2 B^{[2]}. \tag{27}$$

The nonnegative coefficients sum to one, so $H(t)$ is positive definite on the entire closed interval. In particular, $C_1 + tC_2$ is positive definite. The middle matrix in (26) is a Bernstein control matrix, not the midpoint sample $H(\delta/2)$; in general it omits the term $\delta^2 H_2/4$ present in that sample. For $0 < t \le \delta$, put $S = \operatorname{diag}(I, (t/\delta)I)$. Direct block multiplication gives

$$T^T Q(t) T = SH(t)S + \operatorname{diag}\big(0, t(1-t/\delta)(C_1 + tC_2)\big). \tag{28}$$

The first term is positive definite, because S is invertible, and the second term is positive semidefinite. Thus $Q(t) > 0$ for every stated t. Appendix A gives the bases and all reconstruction details, including an independently implemented check of (28) coefficientwise through degree four. No numerical sign decision is used. ∎

3.1 Storage inequalities on actual histories

For $k \ge 1$, let

$$h_k = (Ju_k, Iv_k, \dots, Ju_{k+3}, Iv_{k+3}), \qquad \psi_k = (F_p(u_k), H_p(v_k), \dots, F_p(u_{k+3}), H_p(v_{k+3})),$$

and define

$$V_k(p) = \langle h_k, P(t)h_k\rangle + c(t)^T \psi_k. \tag{29}$$

It is finite, although the entries of c have both signs and P need not be positive definite. With w_k the full five-pair history, the matrix and value identities imply exact equalities

$$\begin{aligned} V_k(p) - V_{k+1}(p) &= \langle w_k, D(t)w_k\rangle + \sum_{e\in\mathcal{A}} \lambda_e(t) S_{e,k}, \\ V_k(p) &= \langle h_k, C_8(t)h_k\rangle + \sum_{e\in\mathcal{B}} \mu_e(t) S_{e,k}. \end{aligned} \tag{30}$$

All slacks are nonnegative by Section 2. Each finite real symmetric matrix inequality lifts to Hilbert-valued slots: expand the finitely many vectors in an orthonormal basis of their finite-dimensional span and sum the scalar inequalities for the coordinates.

For the fixed $t > 0$, let $\alpha_t > 0$ and $\beta_t > 0$ be lower spectral bounds for $C_8(t)$ and the leading block $D_8(t)$ of $D(t)$, respectively. The isometries give

$$V_k(p) \ge \alpha_t \sum_{i=0}^{3} \big(\|u_{k+i}\|^2 + \|v_{k+i}\|^2\big) \ge 0. \tag{31}$$

For arbitrary five-pair slots $w = (w_0, \dots, w_4)$, subtract the last pair from every pair. The identity $DZ = 0$ yields

$$\langle w, Dw\rangle = \langle (w_0 - w_4, \dots, w_3 - w_4), D_8(w_0 - w_4, \dots, w_3 - w_4)\rangle \ge \beta_t \sum_{i=0}^{3} \|w_i - w_4\|^2. \tag{32}$$

In particular, the $i = 3$ summand controls the newest increment, so

$$V_k(p) - V_{k+1}(p) \geq \beta_t\big(\|u_{k+4} - u_{k+3}\|^2 + \|v_{k+4} - v_{k+3}\|^2\big). \tag{33}$$

There is no required positive lower bound for α_t or β_t as $t \downarrow 0$. They are used only at the one fixed parameter selected in (8). The endpoint certificate itself is semicoercive.

4 From the storage to full weak convergence

4.1 Boundedness and square summability

By (31)–(33), the sequence $V_k(p)$ decreases to a finite nonnegative limit. Equation (31) bounds all positive-index iterates. Telescoping (33) proves square summability of $\Delta u_n, \Delta v_n$ for $n \geq 5$. The finitely many omitted increments are finite Hilbert-space vectors and can be added. Thus, with $\Delta u_n = u_n - u_{n-1}$ and $\Delta v_n = v_n - v_{n-1}$,

$$\sum_{n=1}^{\infty}(\|\Delta u_n\|^2 + \|\Delta v_n\|^2) < \infty, \qquad \Delta u_n \to 0, \quad \Delta v_n \to 0. \tag{34}$$

This also proves boundedness of the sequence including its arbitrary initial pair. Starting the history at $k = 1$ is what avoids any assumption that $f(x_0)$ or $h(y_0)$ is finite.

4.2 The signed value terms vanish

The upper anchor inequalities (14) and the literal inclusions (12) give

$$\begin{aligned} 0 &\leq F_p(u_n) + H_p(v_n) \leq \langle a_n, u_n\rangle + \langle b_n, v_n\rangle \\ &= -\frac{1}{r}\langle \Delta u_n, u_n\rangle - \langle \Delta v_n, v_n\rangle + \langle Ku_n, \Delta v_n\rangle + \frac{1}{8}\langle K\Delta u_n, v_n\rangle \longrightarrow 0. \end{aligned} \tag{35}$$

The equality follows by collecting the two cross terms: $-\langle Ku_n, v_{n-1}\rangle + \langle Ku_n, v_n\rangle = \langle Ku_n, \Delta v_n\rangle$. Boundedness and (34) justify the limit. Since the two shifted values are separately nonnegative,

$$F_p(u_n) \to 0, \qquad H_p(v_n) \to 0. \tag{36}$$

Consequently, $c(t)^T\psi_k \to 0$ for each fixed certificate parameter and every KKT anchor p. The signed terms have not been dropped before proving this limit. No continuity of a convex function along weakly convergent sequences is being assumed.

4.3 A limiting form that separates KKT points

Every finite-lag difference tends to zero by (34). Boundedness and the finiteness of P therefore imply that replacing all four historical pairs by the first current pair changes the quadratic storage by $o(1)$. From (36),

$$V_k(p) = q(z_k - p) + o(1), \quad z_k = (X_k, Y_k), \tag{37}$$

where, with $M = Z_8^T P(t) Z_8$,

$$q(a,b) = M_{11}\|a\|^2 + 2M_{12}\langle Ka, b\rangle + M_{22}\|b\|^2. \tag{38}$$

In particular, $q(z_k - p)$ has a finite limit for every normalized KKT point. The matrix M is independent of that point. By (22) and $C_8 > 0$,

$$M_{11} > 0, \qquad M_{22} > 0. \tag{39}$$

We do not need M to be positive definite on $\mathbb{R}^2$, or the quadratic form q to be positive on all of $\mathcal{H} \times \mathcal{G}$.

Indeed, let $p = (U, V)$ and $p' = (U', V')$ be normalized KKT pairs. Monotonicity of ∂F and ∂H gives, respectively,

$$-\langle K(U - U'), V - V'\rangle \geq 0, \qquad \langle K(U - U'), V - V'\rangle \geq 0.$$

Thus the cross pairing is zero, and

$$q(p - p') = M_{11}\|U - U'\|^2 + M_{22}\|V - V'\|^2 > 0 \quad \text{if } p \neq p'. \tag{40}$$

This is the precise restricted positivity needed for convergence.

4.4 Every weak cluster point is a KKT point

For completeness we give a direct subgradient proof, without relying on a monotone-operator sum theorem. The normalized residuals

$$e_n = -\Delta X_n/r + K^*\Delta Y_n, \qquad d_n = -\Delta Y_n + \tfrac{1}{8}K\Delta X_n \tag{41}$$

tend strongly to zero and satisfy

$$-K^*Y_n + e_n \in \partial F(X_n), \qquad KX_n + d_n \in \partial H(Y_n).$$

Let $z_{n_j} \rightharpoonup \bar{z} = (\bar{X}, \bar{Y})$. Such subsequences exist because the orbit is bounded in a Hilbert space. For one fixed KKT anchor, (11) and (36) show that $F(X_n)$ and $H(Y_n)$ are bounded. They also have global affine lower bounds supplied by that anchor. Weak lower semicontinuity gives $F(\bar{X}), H(\bar{Y}) < +\infty$.

For $A \in \operatorname{dom} F$ and $B \in \operatorname{dom} H$, add the two actual subgradient inequalities and cancel the current-point skew terms. This yields

$$\begin{aligned} F(A) + H(B) \geq F(X_n) + H(Y_n) - \langle Y_n, KA\rangle + \langle KX_n, B\rangle \\ + \langle e_n, A - X_n\rangle + \langle d_n, B - Y_n\rangle, \end{aligned} \tag{42}$$

The residual terms vanish along the subsequence by boundedness. The function $(X, Y) \mapsto F(X) + H(Y)$ is weakly lower semicontinuous; this also follows by subtracting the two

fixed affine lower bounds and adding nonnegative weakly lower-semicontinuous functions. Passing to the lower limit gives

$$F(A) + H(B) \geq F(\bar{X}) + H(\bar{Y}) - \langle \bar{Y}, KA \rangle + \langle K\bar{X}, B \rangle. \tag{43}$$

Taking $B = \bar{Y}$ and then $A = \bar{X}$ proves $-K^*\bar{Y} \in \partial F(\bar{X})$ and $K\bar{X} \in \partial H(\bar{Y})$. Every weak cluster point is therefore KKT.

4.5 Uniqueness and convergence of the full sequence

Let Q denote the bounded selfadjoint operator representing (38) on the product Hilbert space:

$$\mathsf{Q} = \begin{pmatrix} M_{11}I & M_{12}K^* \\ M_{12}K & M_{22}I \end{pmatrix}.$$

For two weak cluster points p, p', both are KKT, and (37) implies that $q(z_k - p) - q(z_k - p')$ has a limit. On the other hand,

$$q(z_k - p) - q(z_k - p') = -2\langle z_k, \mathsf{Q}(p - p') \rangle + q(p) - q(p'). \tag{44}$$

The right side is affine and weakly continuous in z_k. Along subsequences tending weakly to p and p', its limits are $-q(p - p')$ and $q(p - p')$, respectively. They must agree, so $q(p - p') = 0$. Equation (40) forces $p = p'$.

A bounded Hilbert-space sequence with a unique weak cluster point converges weakly to that point. To see this directly, if convergence failed, some continuous linear functional would detect a subsequence staying a fixed positive distance from its value at the proposed limit. That subsequence has a weakly convergent further subsequence, contradicting uniqueness. Undoing (9) proves the boundedness and weak convergence in Theorem 1. The scalar norm factors in that normalization transfer square summability. The original proximal inclusions give (6), which tends strongly to zero by increment decay. Weak and norm convergence coincide in finite dimensions. This proves all strict-range conclusions of the theorem.

5 The endpoint triangle cycle

We give the endpoint obstruction with explicit projection inequalities, so it can be checked independently of the storage certificate. In $\mathbb{R}^2$ use

$$\langle (a, b), (c, d) \rangle_G = ac + 3bd, \quad R = \begin{pmatrix} -1/2 & -3/2 \\ 1/2 & -1/2 \end{pmatrix}, \quad e = (1, 0). \tag{45}$$

Then R is an isometry, $R^3 = I$, and $I + R + R^2 = 0$. Put

$$\nu = \frac{11 + 2\sqrt{91}}{18}, \qquad w = (\nu, -1/2), \qquad c_i = R^i e, \qquad d_i = R^i w / r_\star \quad (i = 0, 1, 2). \tag{46}$$

Thus $d_0 = (9/16, -1/(2r_\star))$, and

$$\frac{3}{2} < \nu < \frac{15}{8}, \qquad 27\nu^2 - 33\nu - \frac{81}{4} = 0, \qquad r_\star = \frac{16\nu}{9}. \tag{47}$$

Let $C = \mathrm{conv}\{c_0, c_1, c_2\}$ and $D = \mathrm{conv}\{d_0, d_1, d_2\}$. Take $f = \iota_C$, $h = \iota_D$, $L = I$, $\tau = r_\star$, and $\sigma = 1$. Both triangles have the origin as barycenter, so $(0, 0)$ is a KKT pair. The identity has norm one in this Hilbert plane, and the product is $r_\star$.

We show that starting at (c_0, d_0) gives the orbit (c_i, d_i), with indices taken modulo three. Rotation invariance reduces the proof to the update $i = 0$ to $i = 1$. The primal projection residual is

$$c_0 - r_\star d_0 - c_1 = e - w - Re = (3/2 - \nu, 0).$$

Its inner products with $c_0 - c_1 = (3/2, -1/2)$ and $c_2 - c_1 = (0, -1)$ are

$$\frac{3}{2}(3/2 - \nu) < 0, \qquad 0, \tag{48}$$

respectively. The product against $c_1 - c_1$ is zero. Thus the residual belongs to $N_C(c_1)$, proving the primal projection.

For the dual residual put $q = d_0 + (9c_1 - c_0)/8 - d_1$. Direct multiplication in the inner product (45) gives

$$r_\star^2 \langle q, d_0 - d_1 \rangle_G = 3(\nu^2 + 3/4) - \frac{r_\star}{8}(15\nu + 6), \tag{49}$$

$$r_\star^2 \langle q, d_2 - d_1 \rangle_G = \frac{3}{2}(\nu^2 + 3/4) - \frac{r_\star}{16}(27\nu - 33/2). \tag{50}$$

The identities in (47) imply

$$24(\nu^2 + 3/4) = r_\star(27\nu - 33/2).$$

Equation (50) is therefore zero, while (49) equals $r_\star(12\nu - 45/2)/8 < 0$. These are all nontrivial vertex tests for $q \in N_D(d_1)$. Hence the dual projection also holds. Applying R proves every subsequent update. The primal vertices are distinct, so the orbit has period exactly three and is not convergent.

The isometry $(a, b) \mapsto (a, \sqrt{3}b)$ converts this example to the usual Euclidean plane if desired. Both exact verifiers check all endpoint normal inequalities directly, with the second using an independently chosen Gram-coordinate representation. A strict universal upper product bound larger than $r_\star$ would include this instance. Combined with the strict-range proof, this establishes the sharpness in Theorem 1. The example is bounded; it does not by itself settle boundedness of all other endpoint instances.

6 Scope of the certificate argument

The positivity proof is uniform over a parameter interval only in the sense of validity at every point. It does not provide uniform positive coercivity constants up to $t = 0$. In

fact, the endpoint descent form annihilates constants, ramps, and a critical three-cycle, while its coercivity form annihilates constants and $Z_8^T P(0) Z_8 = 0$. Those degeneracies explain why the strict proof cannot be evaluated at the endpoint to assert boundedness or convergence.

The proof requires constant positive steps and the precise update order in (2). It establishes no theorem for arbitrary extrapolation, variable steps, outer relaxation, or prediction–correction variants. The conclusion about sharpness is an exact fixed-θ universal convergence statement. It is not a statement that the bilinear model is worst case, nor a consequence of searching for or excluding finitely many periodic orbits. The sufficient direction comes from the full storage and weak-convergence proof; the cycle supplies only the matching necessary obstruction.

7 A sharp threshold on a nondegenerate extrapolation interval

We now allow the extrapolation to vary, while the steps and the two objectives remain fixed. The assumptions on $\mathcal{H}, \mathcal{G}, f, h, L$ and the KKT set are those of (1). The literal update is

$$\begin{aligned} x_{n+1} &= \operatorname{prox}_{\tau f}(x_n - \tau L^* y_n), \\ y_{n+1} &= \operatorname{prox}_{\sigma h}\big(y_n + \sigma L((1+\theta)x_{n+1} - \theta x_n)\big). \end{aligned} \tag{51}$$

Define, for $\theta \geq 0$,

$$S(\theta) = \sqrt{4 + 12\theta + 12\theta^2}, \qquad R_3(\theta) = \frac{S(\theta) + 1 + 3\theta}{(1+\theta)^2}, \tag{52}$$

and put

$$\mathcal{I} = \left[\frac{1480281}{11883719}, \frac{237969}{1901231}\right] = [0.1245637834\ldots, 0.1251657478\ldots]. \tag{53}$$

The exact rational endpoints, rather than their decimal display, define this closed interval.

Theorem 2. *Fix $\theta \in \mathcal{I}$ and positive constants τ, σ. Suppose $\mathcal{Z} \neq \varnothing$. For every initial pair in $\mathcal{H} \times \mathcal{G}$, if*

$$\rho = \tau\sigma\|L\|^2 < R_3(\theta), \tag{54}$$

then (51) *is bounded and converges weakly to a KKT pair. Its increments satisfy* (5)*. For $n \geq 1$,*

$$\left(-\frac{x_n - x_{n-1}}{\tau} + L^*(y_n - y_{n-1}),\ -\frac{y_n - y_{n-1}}{\sigma} + \theta L(x_n - x_{n-1})\right) \tag{55}$$

belongs to $(\partial f(x_n) + L^* y_n) \times (\partial h(y_n) - L x_n)$ and tends strongly to zero. In finite dimensions the whole sequence converges in norm. For each fixed $\theta \in \mathcal{I}$, $R_3(\theta)$ is the sharp universal strict product cap for this class.

The interval contains $1/8$ in its interior, and $R_3(1/8) = r_\star$. Thus the conclusion at $1/8$ is precisely the previously proved Theorem 1; it is not a separate new convergence result. The new sufficient range is a genuine continuum in θ, covering every strict product at every point of that interval. No maximality of $\mathcal{I}$ is claimed. Equality in (54) is excluded even at the two included endpoints of $\mathcal{I}$. The statement gives neither infinite-dimensional strong convergence nor universal endpoint boundedness. The sharpness assertion will follow from the actual triangles in Theorem 3. The sufficient proof below is independent of that periodic construction.

7.1 A continuous exact certificate

Use the rational parametrization

$$\begin{aligned} u &= (1+\theta)R_3(\theta), \qquad R(u) = \frac{-u^2+6u+3}{4}, \\ \theta(u) &= \frac{u^2-2u-3}{-u^2+6u+3}, \qquad J = \left[\frac{3341}{1000}, \frac{1337}{400}\right]. \end{aligned} \tag{56}$$

On J, $R > 0$, $\theta'(u) > 0$ and $\theta(J) = \mathcal{I}$. The identities $(1+\theta)R = u$ and

$$(1+\theta)^2R^2 - 2(1+3\theta)R - 3 = 0 \tag{57}$$

identify R with the positive root (52). Keep $\delta = 1/100000$ and let $r = R(u) - t$, $0 < t \le \delta$. The public rational functions define

$$\begin{aligned} P(u,t) &= \frac{P_0(u) + tP_1(u) + t^2P_2(u)}{r}, \quad c(u,t) = c_0(u) + tc_1(u), \\ \lambda(u,t) &= \lambda_0(u) + t\lambda_1(u), \qquad\qquad \mu(u,t) = \mu_0(u) + t\mu_1(u). \end{aligned} \tag{58}$$

Here P is symmetric 8×8, c has eight entries, and λ, μ have respectively 50 and 32 entries. Their exact interpretation and the complete continuous positivity proof appear in Appendix B.

For clarity we state all the properties needed by the analytic argument. Use the ten slots $(X_0, Y_0, \dots, X_4, Y_4)$, with formal residuals

$$a_i = (X_{i-1} - X_i)/r - Y_{i-1}, \qquad b_i = Y_{i-1} - Y_i + (1+\theta)X_i - \theta X_{i-1} \quad (1 \le i \le 4). \tag{59}$$

Every ordered convexity slack is written $S_e = \langle w, Q_e w\rangle + d_e^T\phi$; the anchor has point, gradient and value zero. Let O, N select respectively the first and last four pairs of a five-pair history. Let Z be the ten-by-two constant-history matrix with every pair block I_2, and let Z_8 be its first eight rows. Define

$$\begin{aligned} D &= O^TPO - N^TPN - \sum_e \lambda_e Q_e, \\ C &= O^TPO - \sum_e \mu_e Q_e. \end{aligned} \tag{60}$$

The value-flow identities are

$$O^T c - N^T c - \sum_e \lambda_e d_e = 0, \qquad O^T c - \sum_e \mu_e d_e = 0. \tag{61}$$

Selectors in this equation act on value slots in the same order as point slots. The final two rows and columns of C are zero. Write D_8, C_8 for the leading eight-by-eight blocks, and $M = Z_8^T P Z_8$.

Lemma 2. *For every $u \in J$ and $0 < t \le \delta$, the data in* (58) *have no poles and satisfy* (61)*, $DZ = 0$, $D_8 > 0$, $C_8 > 0$, and $\lambda, \mu \ge 0$. Moreover,*

$$M_{11} = (Z_8^T C_8 Z_8)_{11} > 0, \qquad M_{22} = (Z_8^T C_8 Z_8)_{22} > 0. \tag{62}$$

At $u_0 = (11 + 2\sqrt{91})/9$ all 288 scalar coefficient slots in (58) *recover the complete old coefficient family* (23)*, including its zero multipliers.*

The proof is the exact reconstruction and Bernstein–Gershgorin argument in Appendix B, not a claim of continuity from an individual strictly subcritical product. Both independent implementations reconstruct the literal residual slacks, including the value identities. The finite algebraic inputs and the interval argument give the entire rectangle $J \times (0, \delta]$.

7.2 General coupling and arbitrary initialization

Fix the parameters of Theorem 2, and choose once

$$t = \min\{\delta, (R_3(\theta) - \rho)/2\}, \qquad r = R_3(\theta) - t. \tag{63}$$

Then $0 < t \le \delta$, $r > \rho \ge 0$. Introduce

$$X = \sqrt{r/\tau}\, x, \qquad Y = y/\sqrt{\sigma}, \qquad K = \sqrt{\tau\sigma/r}\, L, \tag{64}$$

with $F(X) = f(\sqrt{\tau/r}X)$ and $H(Y) = h(\sqrt{\sigma}Y)$. The proximal optimality inclusions and the scalar chain rule give

$$\begin{aligned} (X_{n-1} - X_n)/r - K^* Y_{n-1} &\in \partial F(X_n), \\ Y_{n-1} - Y_n + K((1+\theta)X_n - \theta X_{n-1}) &\in \partial H(Y_n). \end{aligned} \tag{65}$$

The normalized KKT set consists of pairs $p = (U, V)$ with $-K^*V \in \partial F(U)$ and $KU \in \partial H(V)$. Since $\|K\| < 1$, on $\mathcal{E} = \mathcal{G} \oplus \mathcal{H}$ the maps

$$Ju = (Ku, (I - K^*K)^{1/2}u), \qquad Iv = (v, 0) \tag{66}$$

are isometries and satisfy $\langle Ju, Iv\rangle = \langle Ku, v\rangle$. This construction covers $L = 0$ without a separate inversion or closed-range assumption.

For any normalized KKT anchor p, put $u_n = X_n - U$, $v_n = Y_n - V$, and

$$\begin{aligned} F_p(u) &= F(U+u) - F(U) + \langle K^*V, u\rangle, \\ H_p(v) &= H(V+v) - H(V) - \langle KU, v\rangle. \end{aligned} \tag{67}$$

These two functions are separately nonnegative, vanish at zero, and have zero subgradient at zero. Subtracting the KKT affine terms in (65) gives the actual subgradients

$$(u_{n-1} - u_n)/r - K^*v_{n-1}, \qquad v_{n-1} - v_n + K((1+\theta)u_n - \theta u_{n-1}).$$

For a segment starting at $k \geq 1$, substitute $X_i = Ju_{k+i}$, $Y_i = Iv_{k+i}$, and the corresponding values of F_p, H_p into every slack in (59)–(60). Same-variable and cross-variable pairings are preserved by (66). Each formal slack is consequently a genuine convexity inequality. This argument only preserves pairings; it does not assert that the formal gradients are subgradients of extensions on $\mathcal{E}$.

A proximal output of a proper lower-semicontinuous convex function has a finite function value and a subgradient. Thus all positive-index values used here are finite. The earliest position of a segment is used only as a point and value, never as a selected subgradient. Starting at $k = 1$ avoids any assumption that $f(x_0)$ or $h(y_0)$ is finite.

7.3 Coercivity and decay of increments and signed values

Let $h_k = (Ju_k, Iv_k, \ldots, Ju_{k+3}, Iv_{k+3})$, and let ψ_k be its eight shifted values. Define the possibly signed-value storage

$$V_k(p) = \langle h_k, Ph_k\rangle + c^T\psi_k. \tag{68}$$

Let w_k be the corresponding five-pair vector. The exact matrix and value identities imply

$$\begin{aligned} V_k(p) - V_{k+1}(p) &= \langle w_k, Dw_k\rangle + \sum_e \lambda_e S_e, \\ V_k(p) &= \langle h_k, C_8 h_k\rangle + \sum_e \mu_e S_e. \end{aligned} \tag{69}$$

Scalar positive-definite matrix inequalities apply to these Hilbert slots: expand the finite-dimensional span of the finitely many slot vectors in an orthonormal basis and sum the scalar inequalities. Let $\alpha, \beta > 0$ be the smallest eigenvalues of C_8, D_8 for the fixed u, t. Then

$$V_k(p) \geq \alpha \sum_{i=0}^{3} (\|u_{k+i}\|^2 + \|v_{k+i}\|^2). \tag{70}$$

Because $DZ = 0$, subtracting the last pair from all five pairs leaves the descent quadratic unchanged. Positivity of D_8 therefore gives

$$V_k(p) - V_{k+1}(p) \geq \beta \sum_{i=0}^{3} \big(\|u_{k+i} - u_{k+4}\|^2 + \|v_{k+i} - v_{k+4}\|^2\big). \tag{71}$$

It follows that $V_k(p)$ decreases to a finite nonnegative limit, the orbit is bounded, and the increments are square summable. The newest increment is explicitly one of the terms in (71); only finitely many initial increments are omitted. No lower bound uniform as $t \downarrow 0$ is needed or claimed.

Write $\Delta u_n = u_n - u_{n-1}$ and $\Delta v_n = v_n - v_{n-1}$. Upper subgradient inequalities at zero give, with the literal residuals,

$$\begin{aligned} 0 \le F_p(u_n) + H_p(v_n) \le &-\langle \Delta u_n, u_n\rangle/r - \langle \Delta v_n, v_n\rangle \\ &+ \langle Ku_n, \Delta v_n\rangle + \theta\langle K\Delta u_n, v_n\rangle \longrightarrow 0. \end{aligned} \tag{72}$$

Boundedness and increment decay justify the limit term by term. The two function values are separately nonnegative, so each tends to zero. Consequently every signed c-weighted value term in (68) vanishes. Discarding those coefficients by their sign would not be valid.

7.4 The limiting form and closure of the KKT conditions

Finite-lag differences vanish, and the histories are bounded. Hence

$$V_k(p) = q(z_k - p) + o(1), \qquad z_k = (X_k, Y_k), \tag{73}$$

where

$$q(a,b) = M_{11}\|a\|^2 + 2M_{12}\langle Ka, b\rangle + M_{22}\|b\|^2. \tag{74}$$

For every KKT p, $q(z_k - p)$ therefore has a limit. The full form q need not be positive definite. If p, p' are KKT and $(a,b) = p - p'$, monotonicity of ∂F gives $\langle Ka, b\rangle \le 0$, while monotonicity of ∂H gives the reverse inequality. Thus

$$q(p - p') = M_{11}\|a\|^2 + M_{22}\|b\|^2 > 0 \quad \text{when } p \ne p'. \tag{75}$$

This is exactly the required KKT-restricted separation.

To prove that all weak cluster points are KKT, set

$$e_n = -\Delta X_n/r + K^*\Delta Y_n, \qquad d_n = -\Delta Y_n + \theta K\Delta X_n.$$

They tend strongly to zero and satisfy

$$-K^*Y_n + e_n \in \partial F(X_n), \qquad KX_n + d_n \in \partial H(Y_n). \tag{76}$$

Take any weakly convergent subsequence with limit $(\bar{X}, \bar{Y})$. The anchored values in (67) tend to zero, and their affine terms are bounded. Thus $F(X_n), H(Y_n)$ are bounded along the subsequence. Their fixed affine lower bounds and weak lower semicontinuity show $\bar{X} \in \operatorname{dom} F$, $\bar{Y} \in \operatorname{dom} H$. For arbitrary $A \in \operatorname{dom} F$, $B \in \operatorname{dom} H$, add the two subgradient inequalities in (76). The skew current-point terms cancel exactly, yielding

$$\begin{aligned} F(A) + H(B) \ge F(X_n) + H(Y_n) - \langle Y_n, KA\rangle + \langle KX_n, B\rangle \\ + \langle e_n, A - X_n\rangle + \langle d_n, B - Y_n\rangle. \end{aligned} \tag{77}$$

The error terms vanish. After subtracting the fixed affine lower bounds, weak lower semicontinuity applies to the sum of the two nonnegative anchored functions. Taking lower limits in (77) gives

$$F(A) + H(B) \geq F(\bar{X}) + H(\bar{Y}) - \langle \bar{Y}, KA \rangle + \langle K\bar{X}, B \rangle.$$

First set $B = \bar{Y}$, then $A = \bar{X}$. These two choices prove $-K^*\bar{Y} \in \partial F(\bar{X})$ and $K\bar{X} \in \partial H(\bar{Y})$ separately. No weak–weak closedness of an individual subdifferential graph has been assumed.

7.5 Convergence of the whole sequence and sharpness

Let $\mathcal{Q}$ be the bounded selfadjoint operator representing q on $\mathcal{H} \times \mathcal{G}$. If p, p' are weak cluster points, both are KKT, and the difference

$$q(z_k - p) - q(z_k - p') = -2\langle z_k, \mathcal{Q}(p - p') \rangle + q(p) - q(p') \tag{78}$$

has a limit. Its right side is affine and weakly continuous in z_k. Along subsequences tending to p and p', its limits are respectively $-q(p - p')$ and $q(p - p')$. Therefore $q(p - p') = 0$, and (75) proves $p = p'$.

Bounded sequences in a Hilbert space have weakly convergent subsequences, also in the nonseparable case. If the whole sequence failed to converge weakly to its unique weak cluster point, one continuous linear functional would separate a subsequence from that point. A further weakly convergent subsequence contradicts uniqueness. This proves whole-sequence weak convergence. Undoing the fixed normalization preserves boundedness, weak convergence and square summability. The original residuals (55) follow directly from the original proximal inclusions, and tend strongly to zero. Finite-dimensional weak convergence is norm convergence.

Theorem 3 below supplies, for each fixed $\theta \in \mathcal{I}$, an actual nonconvergent instance with product exactly $R_3(\theta)$ and with a unique KKT pair. Any proposed larger universal strict cap would include that instance. Together with the sufficient proof just completed, this proves the last assertion of Theorem 2.

8 The exact period three onset for every nonnegative extrapolation

The next result holds on the much larger range $\theta \geq 0$. Its conclusion concerns actual periodic orbits with two fixed convex objectives. It is logically distinct from the convergence theorem.

Theorem 3. *Let $\theta \geq 0$, $r > 0$, and let f, h be fixed proper lower-semicontinuous convex functions on an arbitrary real Hilbert space. Consider*

$$x_{n+1} = \operatorname{prox}_{rf}(x_n - ry_n), \qquad y_{n+1} = \operatorname{prox}_h(y_n + (1+\theta)x_{n+1} - \theta x_n). \tag{79}$$

If $r < R_3(\theta)$, every three-periodic orbit is constant, without assuming KKT existence. For every fixed $\theta \geq 0$, there are two fixed nondegenerate compact triangles in a Hilbert plane whose indicator objectives have the same actual least-period-three orbit for every $r \geq R_3(\theta)$. The KKT pair is uniquely $(0, 0)$. The examples embed in every real Hilbert space of dimension at least two. Thus the existence set for nonconstant three-periodic orbits across Hilbert spaces, with or without a KKT requirement, is exactly $[R_3(\theta), \infty)$.

The exclusion here is stated for $L = I$, $\sigma = 1$, $\tau = r$. Theorem 2, in contrast, allows arbitrary bounded L. The absence of period three by itself implies no convergence conclusion below this curve and excludes neither other periods nor aperiodic nonconvergence. In particular, $R_3(0) = 3$ is not a small-step convergence statement at zero extrapolation.

8.1 Cyclic interpolation of actual proximal data

Indices in this subsection are modulo three. Every phase of an actual periodic orbit is a proximal output, including phase zero through the update from phase two. Therefore

$$\begin{aligned} p_i &= (x_{i-1} - x_i)/r - y_{i-1} \in \partial f(x_i), \\ q_i &= y_{i-1} - y_i + (1+\theta)x_i - \theta x_{i-1} \in \partial h(y_i). \end{aligned} \tag{80}$$

For each $k = 1, 2$, summing the three subgradient inequalities cancels the function values and gives

$$\sum_i \langle p_i, x_{i+k} - x_i \rangle \leq 0, \qquad \sum_i \langle q_i, y_{i+k} - y_i \rangle \leq 0. \tag{81}$$

The fixed-objective requirement is used precisely in this telescoping. These are cyclic-monotonicity inequalities, not merely pairwise graph monotonicity.

If the primal triple is constant, the dual inequality with $k = 1$ becomes

$$\sum_i \langle y_{i-1} - y_i, y_{i+1} - y_i \rangle = \tfrac{1}{2} \sum_i \|y_i - y_{i-1}\|^2 \leq 0,$$

so the dual triple is also constant. For a nonconstant orbit let $\kappa^2 = \sum_i \|x_i - \bar{x}\|^2 > 0$, where bars denote the three-point means, and set in the product Hilbert space

$$X = (x_0 - \bar{x}, x_1 - \bar{x}, x_2 - \bar{x})/\kappa, \qquad Y = (y_0 - \bar{y}, y_1 - \bar{y}, y_2 - \bar{y})/\kappa.$$

Let $T(u_0, u_1, u_2) = (u_1, u_2, u_0)$. On the centered subspace, $T^2 + T + I = 0$ and T is an isometry. Consequently $\mathcal{J} = (I + 2T)/\sqrt{3}$ satisfies $\mathcal{J}^* = -\mathcal{J}$, $\mathcal{J}^2 = -I$, and is an isometry. Since $X, \mathcal{J}X$ are orthonormal, the quantities

$$a = \langle X, Y \rangle, \qquad b = \langle \mathcal{J}X, Y \rangle/\sqrt{3}, \qquad v = \|Y\|^2 \tag{82}$$

satisfy $v \geq a^2 + 3b^2$.

The constant residual terms created by centering cancel against the summed differences. Divide (81) by κ^2 and use

$$P = (T^2X - X)/r - T^2Y, \qquad Q = T^2Y - Y + (1+\theta)X - \theta T^2X.$$

Substitution of $T = -I/2 + \sqrt{3}\mathcal{J}/2$ gives the four left sides

$$\begin{aligned} \langle P, TX - X\rangle &= 3b + 3/(2r), \\ \langle P, T^2X - X\rangle &= -3a/2 + 3b/2 + 3/r, \\ \langle Q, TY - Y\rangle &= \tfrac{3}{2}[v - (1+\theta)a - (1+3\theta)b], \\ \langle Q, T^2Y - Y\rangle &= 3[v - (\theta + \tfrac{1}{2})a + b/2]. \end{aligned} \tag{83}$$

Thus every nonconstant orbit yields

$$\begin{gathered} b \leq -1/(2r), \qquad a - b \geq 2/r, \\ a^2 + 3b^2 \leq v \leq (1+\theta)a + (1+3\theta)b, \qquad v \leq (\theta + \tfrac{1}{2})a - b/2. \end{gathered} \tag{84}$$

No KKT point, finite-dimensionality, differentiability, or finite initial domain assumption was needed.

One can also realize the centering as a literal direct-sum symmetrization. On the product space define

$$\begin{aligned} F_0(u) &= \kappa^{-2} \sum_j f(\kappa u_j + \bar{x}) + \kappa^{-1} \sum_j \langle \bar{y}, u_j\rangle, \\ H_0(w) &= \kappa^{-2} \sum_j h(\kappa w_j + \bar{y}) - \kappa^{-1} \sum_j \langle \bar{x}, w_j\rangle. \end{aligned} \tag{85}$$

The states (T^iX, T^iY) satisfy the same literal map with F_0, H_0: the selected subgradients are coordinatewise $(p_j + \bar{y})/\kappa$ and $(q_j - \bar{x})/\kappa$. Their values are invariant under cyclic permutation. The affine tilts are essential; the original means were never assumed to be KKT. The direct summed proof above does not require this additional construction.

8.2 Exact exclusion and planar attainment

The first primal and first dual inequalities already imply

$$\left(a - \frac{1+\theta}{2}\right)^2 + 3\left(b - \frac{1+3\theta}{6}\right)^2 \leq \frac{S(\theta)^2}{12}.$$

Therefore $b \geq b_- = (1 + 3\theta - S(\theta))/6$. Since $S(\theta)^2 - (1+3\theta)^2 = 3(1+\theta)^2 > 0$, one has $b_- < 0$. Combining $b_- \leq b \leq -1/(2r)$ yields

$$r \geq -\frac{1}{2b_-} = \frac{3}{S(\theta) - 1 - 3\theta} = R_3(\theta). \tag{86}$$

This proves the exclusion directly from actual convex interpolation.

For attainment, use $\mathbb{R}^2$ with inner product $\langle u, v\rangle_M = u_1v_1 + 3u_2v_2$ and the isometry

$$A = \begin{pmatrix} -1/2 & -3/2 \\ 1/2 & -1/2 \end{pmatrix}, \qquad A^3 = I, \qquad I + A + A^2 = 0. \tag{87}$$

Write $r_0 = R_3(\theta)$, $a_0 = (1+\theta)/2$, $b_0 = b_- = -1/(2r_0)$ and set

$$\begin{aligned} c_i &= A^i(1,0), & d_i &= A^i(a_0, b_0), \\ C_\theta &= \operatorname{conv}\{c_0, c_1, c_2\}, & D_\theta &= \operatorname{conv}\{d_0, d_1, d_2\}. \end{aligned} \tag{88}$$

Both triangles are nondegenerate, their centroids are zero, and zero is interior to each. Let $f = \iota_{C_\theta}$, $h = \iota_{D_\theta}$. At $r = r_0$, rotational invariance reduces all normal-cone vertex tests for (80) to (83) with $a = a_0, b = b_0, v = a_0^2 + 3b_0^2$. The first and third tests are zero by (57). The other two tests are nonpositive exactly when

$$a_0 + 3b_0 \geq 0, \qquad a_0 + 3(1+2\theta)b_0 \leq 0. \tag{89}$$

These inequalities hold for every $\theta \geq 0$. Indeed all quantities being compared are positive before squaring, and

$$\begin{aligned} (2+4\theta)^2 - S(\theta)^2 &= 4\theta(1+\theta), \\ (1+2\theta)^2 S(\theta)^2 - (2+6\theta+6\theta^2)^2 &= 4\theta(1+\theta)(3\theta^2+3\theta+1). \end{aligned} \tag{90}$$

The first identity gives the first comparison in (89), and the second gives the second. Hence every residual is in the required normal cone. The projection optimality characterization proves each actual update, not just an interpolation relaxation. The three c_i are distinct, so the least period is exactly three; projections are single valued.

The linear isometry $(u_1, u_2) \mapsto (u_1, \sqrt{3}u_2)$ converts the construction to the ordinary Euclidean plane. It leaves identity coupling and the product r_0 unchanged.

8.3 One pair of triangles for the whole upper ray

For any $r \geq r_0$, retain the same two sets and the same states in (88). The dual normal tests do not depend on r. The primal tests become

$$3b_0 + 3/(2r), \qquad -3a_0/2 + 3b_0/2 + 3/r.$$

They are nonincreasing in r and were nonpositive at r_0. Consequently the same orbit persists simultaneously for every $r \geq r_0$; neither objective is changed with the step product.

The pair $(0,0)$ is KKT because zero is interior to both triangles. If $-y \in N_{C_\theta}(x)$ and $x \in N_{D_\theta}(y)$, tests against zero give respectively $\langle x, y\rangle \leq 0$ and $\langle x, y\rangle \geq 0$. If $y \neq 0$, the first inequality is strict because zero is interior to C_θ, a contradiction. Thus $y = 0$, and $N_{D_\theta}(0) = \{0\}$ forces $x = 0$. After embedding into a larger Hilbert space and retaining

indicators of the planar sets, every KKT coordinate still belongs to its planar domain. Taking planar projections of the normal inclusions gives the same uniqueness conclusion. This proves Theorem 3.

The feasible system (84) is also an exact existential characterization across Hilbert spaces. Any feasible point can have v replaced by a^2+3b^2, making both upper bounds only easier to satisfy. The planar orbits $A^i(1,0)$ and $A^i(a,b)$ then satisfy all four vertex tests for the indicators of their convex hulls. Since $b<0$, the dual triangle is nondegenerate. Thus no unintegrable monotone graph or uncontrolled Gram-rank relaxation is used.

8.4 Exclusion for arbitrary bounded coupling

The identity-coupling lower bound has a direct extension to distinct Hilbert spaces. The following proof uses the actual cross-space residuals; it does not substitute a general operator into the planar construction.

Theorem 4. *Let $\theta \geq 0$, let $L : \mathcal{H} \to \mathcal{G}$ be bounded and linear, and let $\tau, \sigma > 0$. For fixed proper lower-semicontinuous convex f, h, any nonconstant actual three-periodic orbit of* (51) *requires*

$$\rho = \tau\sigma\|L\|^2 \geq R_3(\theta). \tag{91}$$

No KKT existence hypothesis is needed. At equality its centered primal points necessarily satisfy

$$(\|L\|^2 I - L^*L)(x_i - \bar{x}) = 0 \qquad (i = 0, 1, 2). \tag{92}$$

In particular, if L does not attain its norm on a nonzero vector, there is no nonconstant endpoint three-cycle. The bound is sharp over the whole class, including the subclass with KKT existence, but does not assert attainability for every prescribed operator L.

Set $u = x/\sqrt{\tau}$, $v = y/\sqrt{\sigma}$, $A = \sqrt{\tau\sigma}L$, $F_0(u) = f(\sqrt{\tau}u)$ and $H_0(v) = h(\sqrt{\sigma}v)$. The change of variables in the two proximal minimizations gives the unit-step map

$$u^+ = \operatorname{prox}_{F_0}(u - A^*v), \qquad v^+ = \operatorname{prox}_{H_0}(v + A((1+\theta)u^+ - \theta u)),$$

with $\|A\|^2 = \rho$. At each cyclic phase the selected subgradients are

$$P_i = u_{i-1} - u_i - A^*v_{i-1}, \qquad Q_i = v_{i-1} - v_i + A((1+\theta)u_i - \theta u_{i-1}).$$

All phases are actual proximal outputs, and both orientations of the three-point subgradient inequalities telescope as in (81). If the primal triple is constant, the constant forcing in the first dual cyclic sum cancels, leaving half the sum of squared dual increments. The dual triple is then constant as well.

For a nonconstant orbit set $\kappa^2 = \sum_i \|u_i - \bar{u}\|^2 > 0$ and

$$X = (u_i - \bar{u})_{i=0}^2/\kappa \in \mathcal{H}^3, \qquad Y = (v_i - \bar{v})_{i=0}^2/\kappa \in \mathcal{G}^3.$$

Use separate cyclic shifts T_H, T_G and their complex structures $\mathcal{J}_H, \mathcal{J}_G$ on the respective centered spaces. Let A_3 act componentwise. Then $T_G A_3 = A_3 T_H$ and $\mathcal{J}_G A_3 = A_3 \mathcal{J}_H$; no identification of the two Hilbert spaces is involved. Constant tuples caused by centering pair to zero with all cyclic differences. The normalized residual tuples are thus, for purposes of those sums,

$$P = T_H^2 X - X - A_3^* T_G^2 Y, \qquad Q = T_G^2 Y - Y + (1+\theta) A_3 X - \theta T_G^2 A_3 X.$$

Put

$$Z = A_3 X, \quad w = \|Z\|^2, \quad a = \langle Z, Y \rangle, \quad b = \langle \mathcal{J}_G Z, Y \rangle / \sqrt{3}, \quad V = \|Y\|^2. \tag{93}$$

Expanding $T = -I/2 + \sqrt{3}\mathcal{J}/2$ in each space gives

$$\begin{aligned} \langle P, (T_H - I)X \rangle &= 3/2 + 3b, \\ \langle P, (T_H^2 - I)X \rangle &= 3 - 3a/2 + 3b/2, \\ \langle Q, (T_G - I)Y \rangle &= \tfrac{3}{2}[V - (1+\theta)a - (1+3\theta)b], \\ \langle Q, (T_G^2 - I)Y \rangle &= 3[V - (\theta + \tfrac{1}{2})a + b/2]. \end{aligned} \tag{94}$$

For example the first primal cross term is $-\langle T_G^2 Y, (T_G - I)Z \rangle = -\langle Y, (T_G^2 - T_G)Z \rangle = 3b$; the uncoupled term is $3\|X\|^2/2 = 3/2$. Hence cyclic interpolation implies

$$b \le -1/2, \quad a - b \ge 2, \quad V \le (1+\theta)a + (1+3\theta)b, \quad V \le (\theta + \tfrac{1}{2})a - b/2. \tag{95}$$

If $w = 0$, then $Z = 0$ and $b = 0$, contradicting the first inequality. Thus $w > 0$ before any division by w; this also excludes $L = 0$ for a nonconstant cycle.

The two vectors $Z, \mathcal{J}_G Z$ are orthogonal and have squared norm w. Bessel's inequality and the operator norm bound give

$$a^2 + 3b^2 \le wV, \qquad w = \sum_i \|AX_i\|^2 \le \|A\|^2 \sum_i \|X_i\|^2 = \rho. \tag{96}$$

Combining the first bound with the first dual inequality and completing squares yields

$$\left(a - \frac{w(1+\theta)}{2}\right)^2 + 3\left(b - \frac{w(1+3\theta)}{6}\right)^2 \le \frac{w^2 S(\theta)^2}{12}.$$

The exact minimum of b over this ellipse is wb_-, with $b_- = (1 + 3\theta - S(\theta))/6 < 0$. Thus $wb_- \le b \le -1/2$, implying $w \ge -1/(2b_-) = R_3(\theta)$. Together with $w \le \rho$ this proves (91). No norm maximizer or compactness of L was used.

If $\rho = R_3(\theta)$, equality forces $w = \rho$. The bounded operator $D_0 = \rho I - A^* A$ is positive semidefinite, and

$$0 = \rho - w = \sum_i \langle D_0 X_i, X_i \rangle.$$

Every term is nonnegative, so its positive square root satisfies $D_0^{1/2} X_i = 0$, hence $D_0 X_i = 0$. Since $D_0 = \tau\sigma(\|L\|^2 I - L^* L)$ and $X_i = (x_i - \bar{x})/(\sqrt{\tau}\kappa)$, this is (92). At least one

centered primal point is nonzero, and on that vector L attains its norm. This proves the endpoint exclusion under norm nonattainment, including in infinite-dimensional and nonseparable spaces.

The identity triangles of Theorem 3 attain every product $\rho \geq R_3(\theta)$ within the full class. They prove the stated classwise sharpness, while (92) explains why it need not be attainable by each fixed L. Neither assertion entails convergence or boundedness at the endpoint, or excludes another period.

8.5 Comparison with a period two obstruction

For $f = h = 0$ in one dimension, the identity-coupled update matrix is

$$B = \begin{pmatrix} 1 & -r \\ 1 & 1 - (1+\theta)r \end{pmatrix}.$$

At $r_2(\theta) = 4/(1+2\theta)$ it has eigenvalue -1, hence a genuine two-cycle on its nonzero -1 eigenspace. This is the bilinear mechanism in [3, Section 7]. Substitution into the positive-root polynomial (57) gives

$$(1+\theta)^2 r_2^2 - 2(1+3\theta)r_2 - 3 = -\frac{44\theta^2 + 20\theta - 5}{(1+2\theta)^2}.$$

The two obstruction products cross at

$$\theta_{\text{cross}} = \frac{4\sqrt{5} - 5}{22} = 0.1792850868\ldots. \tag{97}$$

Below this value $R_3 < r_2$, and above it $r_2 < R_3$. This comparison does not assert that those mechanisms exhaust all obstructions. At $\theta = 0$ and $0 < r < 4$, the free bilinear matrix has determinant one and trace $2 - r \in (-2, 2)$, so its nonreal eigenvalues have modulus one and it already admits nonconvergent initial states. Thus the value $R_3(0) = 3$ cannot be interpreted as a sufficient convergence threshold.

9 Methodological context and the scope of the extension

The original fixed-1/8 proof and its full algebraic data remain in Sections 1–6 and Appendix A. The continuum certificate adds a sufficient result for general coupling on (53); the independent cyclic argument adds an all-θ classification for identity coupling and the general-coupling exclusion of Theorem 4. Only on the certified interval do the sufficient and necessary bounds combine into a sharp universal convergence threshold.

Convex interpolation and quadratic-plus-value certificates are established tools; see [4, 5]. The converse in [5, Theorem 2] concerns a prescribed certificate class under its

stated feasibility and dimension assumptions; it is not a converse giving an actual nonconvergent orbit whenever that class fails. In [4, Figure 2], the horizontal variable is the equal step $\tau = \sigma$, so comparison with our product requires squaring it. Its $(h, \alpha) = (2, 0)$ history/overlap notation is not interchanged here with the different history indexing discussed in [2]. Numerical search boundaries are not used as sign certificates or as nonconvergence evidence.

Periodic phase constructions also have prior forms. For instance, [6] constructs periodic counterexamples for a scalar feedback interconnection using a frequency/phase condition and scalar monotone interpolation. That mechanism does not by itself prove our necessary threshold for two arbitrary Hilbert convex subdifferentials. Here both orientations of cyclic convex interpolation and every triangle normal test are established explicitly. No priority claim for rotation, periodic constructions, longer memory, or interpolation methodology is made. The specific mathematical extension is the stated exact certificate and its quantifiers, together with the actual periodic classification.

10 Additional sharp regions and a period five obstruction

The convergence proof and the periodic exclusions have different roles. The former requires a positive storage certificate; the latter controls only the specified periods. We first enlarge the sufficient region by verifying the same certificate on a wider interval. We then show, with two fixed convex indicator functions, that a higher period can occur below both of the obstruction curves considered above.

10.1 A wider interval for the same certificate

Define

$$\mathcal{I}_{\mathrm{w}} = \left[\frac{764643}{6190357}, \frac{776347}{6188653}\right], \qquad J_{\mathrm{w}} = \left[\frac{4173}{1250}, \frac{4179}{1250}\right]. \tag{98}$$

The parametrization (56) maps J_{w} increasingly onto $\mathcal{I}_{\mathrm{w}}$. Numerically,

$$\mathcal{I}_{\mathrm{w}} = [0.1235216321\ldots, 0.1254468460\ldots].$$

Its width is approximately 3.1982 times that of $\mathcal{I}$.

Theorem 5. *The full conclusion of Theorem 2 holds for every $\theta \in \mathcal{I}_{\mathrm{w}}$, with the same assumptions on the real Hilbert spaces, fixed proper lower-semicontinuous convex functions, bounded coupling, nonempty KKT set, positive constant step sizes and arbitrary initial pair. In particular, $R_3(\theta)$ is the sharp universal strict product cap throughout this interval.*

Proof. Keep every coefficient in (58) and $\delta = 1/100000$ unchanged. The exact identities in (60)–(61) are rational identities and remain valid wherever their denominators do not vanish. The replay in Appendix C proves, on the single full cell J_{w}, nonvanishing denominators, nonnegative multipliers, both positive metric diagonals, nonsingular kernel complements and positive definiteness of all six desingularized Bernstein controls. Hence every property of Lemma 2 holds on $J_{\mathrm{w}} \times (0, \delta]$. Its coefficientwise recovery at $u_0 = (11 + 2\sqrt{91})/9$ is unchanged.

For each strict product $\rho < R_3(\theta)$ choose t and r by (63). The normalization, contraction embedding, shifted convexity slacks and Hilbert-valued lifting in Section 7.2 therefore apply without any new restriction. They give the coercive, decreasing history storage and square-summable increments by (70)–(71). Separate nonnegative anchored function values tend to zero by (72), so the limiting quadratic form has the two positive diagonal entries needed on differences of KKT pairs. The residual and weak closure proof in Section 7.4, followed by (78), gives whole-sequence weak convergence. This includes $L = 0$ and arbitrary initialization. The residual and finite-dimensional conclusions follow as there. Finally, Theorem 3 supplies the actual endpoint triangle orbit at every θ in this interval, proving sharpness of the strict cap. ∎

The improvement changes only the rational congruence preconditioners used to prove positivity. It does not alter the storage section, its support, its memory or the algorithm. Conversely, the exact negative directions recorded in Appendix C show that this particular section fails its descent positivity test at two nearby parameter values. Such a failure is neither a nonconvergent orbit nor a restriction on other certificates.

10.2 A different certificate near one seventh

The free coordinates can instead be chosen to certify a disjoint component containing $1/7$. Define

$$\mathcal{I}_{\mathrm{s}} = \left[\frac{4161}{29639}, \frac{5707}{39493}\right], \qquad J_{\mathrm{s}} = \left[\frac{169}{50}, \frac{339}{100}\right]. \tag{99}$$

The same rational parametrization maps J_{s} increasingly onto $\mathcal{I}_{\mathrm{s}}$. Numerically,

$$\mathcal{I}_{\mathrm{s}} = [0.1403893518\ldots, 0.1445066214\ldots].$$

Here $\theta = 1/7$ corresponds to $u = (5+\sqrt{73})/4$, strictly between the two rational endpoints.

Theorem 6. *The full conclusion of Theorem 2, with all of its Hilbert-space, objective, KKT, step-size and initialization assumptions, holds for every $\theta \in \mathcal{I}_{\mathrm{s}}$. The sharp universal strict product cap is again $R_3(\theta)$.*

Proof. Use the rational coefficients in `theta_shifted/linear/family.json`, with the same $\delta = 1/100000$, storage form (58), literal residuals (59), edge order and value identities. These coefficients are a different section of the algebraic identities. The continuous

exact replay in Appendix C.2 proves every certificate property needed in the proof of Theorem 5: the two strictly positive reduced forms, nonnegative slack multipliers, constant-history annihilation, and the two positive KKT-separating metric diagonals, on the whole rectangle $J_{\mathrm{s}} \times (0, \delta]$.

For completeness, the use of the certificate does not depend on its particular free coordinates. The fixed choice (63) normalizes an arbitrary bounded coupling to a contraction; the isometries (66) realize exactly every primal, dual and mixed scalar pairing. Starting at positive-index histories makes all shifted function values finite. The value-flow identities and positive reduced forms give a nonnegative decreasing storage and summable squared increments. The separately nonnegative anchored function values vanish by (72), including every signed value contribution to storage. The limiting form consequently exists relative to every KKT anchor, with strict separation of KKT differences by (75). The direct summed-subgradient closure argument (77) places all weak cluster points in the KKT set, and the affine polarization argument (78) makes them identical. This proves all the convergence and residual conclusions, for arbitrary initialization and also for $L = 0$. The endpoint triangles of Theorem 3 give sharpness at each fixed θ. ∎

This section is not required to recover the old 288 coefficients at $\theta = 1/8$; that recovery remains a property of the unchanged section and is still verified there. The two sufficient components $\mathcal{I}_{\mathrm{w}}$ and $\mathcal{I}_{\mathrm{s}}$ do not overlap. No conclusion is drawn for the gap between them, for the product endpoint, or for a maximal extrapolation domain. In particular, a failure of the first section outside its range does not obstruct the second one.

At the distinguished rational extrapolation, the cap is

$$R_3(1/7) = \frac{7(5 + \sqrt{73})}{32}. \tag{100}$$

A separate fixed-parameter certificate gives one additional isolated point, without joining either of the preceding intervals.

Corollary 10.1. *At $\theta = 11/74$, under the Hilbert-space, objective, KKT, positive-step and initialization assumptions of Theorem 2, its full convergence conclusion holds whenever $\tau\sigma\|L\|^2 < 74/25$. This universal strict cap is sharp.*

Proof. The different fixed-anchor coefficients in `theta_shifted/family.json` satisfy the same literal matrix, value-flow, kernel and metric identities at $u = 17/5$. Exact rational LDL^T pivots of the six desingularized Bernstein controls are positive, and all multiplier endpoints are nonnegative, for $0 < t \leq 1/100000$. Appendix C.2 specifies their independent replay. Thus the proof of Theorem 6 applies at this fixed u. The parametrization gives $\theta = 11/74$ and $R = 74/25$ exactly, and the endpoint triangle construction proves sharpness. No extrapolation continuum around this point or between it and $\mathcal{I}_{\mathrm{s}}$ is asserted. ∎

10.3 Five phase convex interpolation

For the construction below take identity coupling, $\tau = r > 0$ and $\sigma = 1$ on one real Hilbert space. All phase indices in this subsection are interpreted modulo five. Set $\phi_1 = 2\pi/5$, $\phi_2 = 4\pi/5$, and, for $1 \le j \le 4$, define

$$\begin{aligned} D_{kj} &= \cos((j+1)\phi_k) - \cos\phi_k - \cos(j\phi_k) + 1, \\ Z_{kj} &= e^{\mathrm{i}\phi_k}(e^{\mathrm{i}j\phi_k} - 1), \\ W_{kj} &= ((1+\theta) - \theta e^{\mathrm{i}\phi_k})(e^{\mathrm{i}j\phi_k} - 1). \end{aligned} \tag{101}$$

For real moments A_k, B_k, c_k, d_k, put

$$\begin{aligned} P_j &= \sum_{k=1}^{2} \left(\frac{D_{kj} A_k}{r} - \operatorname{Re} Z_{kj}\, c_k - \operatorname{Im} Z_{kj}\, d_k \right), \\ H_j &= \sum_{k=1}^{2} \left(D_{kj} B_k + \operatorname{Re} W_{kj}\, c_k - \operatorname{Im} W_{kj}\, d_k \right). \end{aligned} \tag{102}$$

Lemma 3. *An actual nonconstant five-periodic orbit for two fixed proper lower semicontinuous convex functions yields moments satisfying*

$$A_k, B_k \ge 0, \qquad c_k^2 + d_k^2 \le A_k B_k, \qquad A_1 + A_2 = 1, \qquad P_j, H_j \le 0 \quad (1 \le j \le 4). \tag{103}$$

Conversely, any moments satisfying (103) *admit a least-period-five realization for two fixed indicator functions in a Euclidean space of dimension at most eight. This is an existential realization; the functions need not be prescribed in advance.*

Proof. Let F be the forward shift on the five-phase temporal space, equipped with the phase-average inner product. The literal residuals are

$$p = (F^{-1} - I)x/r - F^{-1}y, \qquad q = (F^{-1} - I)y + ((1+\theta)I - \theta F^{-1})x. \tag{104}$$

Convex subgradient inequalities, summed over all phases, telescope the function values and give $\langle p, (F^j - I)x\rangle \le 0$ and $\langle q, (F^j - I)y\rangle \le 0$ for every $1 \le j \le 4$. Centering changes the residuals only by constant sequences, orthogonal to these displacements.

On the centered temporal space the two orthogonal frequency blocks satisfy $F = \cos\phi_k I + \sin\phi_k J_k$, where $J_k^* = -J_k$ and $J_k^2 = -I$. Write $A_k = \|x_k\|^2$, $B_k = \|y_k\|^2$, $c_k = \langle x_k, y_k\rangle$ and $d_k = \langle J_k x_k, y_k\rangle$. The vectors $x_k, J_k x_k$ are orthogonal and have the same norm, so Bessel's inequality gives the cone in (103); if $A_k = 0$, then $c_k = d_k = 0$. Moving the backward shift in (104) to the other factor gives the primal self term $\langle x_k, (F-I)(F^j - I)x_k\rangle/r = D_{kj}A_k/r$ and the primal cross term $-\operatorname{Re} Z_{kj}c_k - \operatorname{Im} Z_{kj}d_k$. The dual self term is $D_{kj}B_k$. Its cross term is $\operatorname{Re} W_{kj}c_k - \operatorname{Im} W_{kj}d_k$, since $\langle x_k, J_k y_k\rangle = -d_k$. This proves all eight tests. If x is constant, the dual test $j = 4$ reduces to $\|(F^{-1} - I)y\|^2 \le 0$, forcing y constant. Otherwise common positive rescaling normalizes $A_1 + A_2$ to one.

For sufficiency, use two orthogonal rotation planes per frequency. When $A_k > 0$, choose within that four-dimensional block

$$x_k = (\sqrt{A_k}, 0, 0, 0), \qquad y_k = \left(\frac{c_k}{\sqrt{A_k}}, \frac{d_k}{\sqrt{A_k}}, \sqrt{B_k - \frac{c_k^2 + d_k^2}{A_k}}, 0 \right),$$

and rotate both planes through ϕ_k at each phase. When $A_k = 0$, use $x_k = 0$ and $y_k = (\sqrt{B_k}, 0)$ on one rotation plane. This retains every Gram slack, including the singular case. Taking the orthogonal sum realizes all moments in dimension at most eight. The common orthogonal rotation makes each residual-to-displaced-vertex pairing independent of phase; its value is exactly the corresponding P_j or H_j in (102). Thus all individual normal inequalities hold for the two fixed convex hulls of the five states. Their indicators give the literal proximal updates. The normalization ensures a nonconstant primal state, and a nonconstant orbit whose period divides the prime number five has least period five. The hulls contain zero because both frequency sums vanish, so $(0, 0)$ is a KKT pair. ∎

The Gram slack in this proof cannot in general be removed: some D_{kj} are negative. The following example saturates both cones by an explicit choice of moments, after which all normal tests are checked afresh.

10.4 A fixed four dimensional example

Let $\mathcal{R}(\phi)$ denote the counterclockwise planar rotation and set $\mathcal{R} = \operatorname{diag}(\mathcal{R}(\phi_1), \mathcal{R}(\phi_2))$. Choose the exact rational data

$$\begin{aligned} (A_1, A_2) &= (106350, 893650)/10^6, \\ (c_1, c_2) &= (16600, 578738)/10^6, \\ (d_1, d_2) &= (-54011, -147051)/10^6, \qquad B_k = (c_k^2 + d_k^2)/A_k. \end{aligned} \tag{105}$$

In Euclidean $\mathbb{R}^4$ define

$$\begin{aligned} x_0 &= (\sqrt{A_1}, 0, \sqrt{A_2}, 0), \\ y_0 &= (c_1/\sqrt{A_1}, d_1/\sqrt{A_1}, c_2/\sqrt{A_2}, d_2/\sqrt{A_2}), \\ x_i &= \mathcal{R}^i x_0, \qquad y_i = \mathcal{R}^i y_0 \quad (0 \le i \le 4), \\ C &= \operatorname{conv}\{x_0, \dots, x_4\}, \qquad D = \operatorname{conv}\{y_0, \dots, y_4\}. \end{aligned} \tag{106}$$

These definitions fix the two objectives $f = \iota_C$ and $h = \iota_D$ independently of the iteration and of the parameters in the next theorem.

Theorem 7. *With $L = I$, $\sigma = 1$ and $\tau = r$, the states* (106) *form an actual least-period-five orbit of* (51) *for every parameter pair in*

$$\frac{179999}{10^6} \le \theta \le \frac{180001}{10^6}, \qquad \frac{293999}{10^5} \le r \le \frac{294001}{10^5}. \tag{107}$$

The fixed objectives are proper closed convex, their KKT set is $\{(0,0)\}$*, and throughout this rectangle*

$$r < R_3(\theta), \qquad r < \frac{4}{1+2\theta}. \tag{108}$$

In particular the central pair is $\theta = 9/50$*,* $r = 147/50$*.*

Proof. All coordinates are explicit radicals, since

$$\cos\phi_1 = \frac{\sqrt{5}-1}{4}, \quad \sin\phi_1 = \frac{\sqrt{10+2\sqrt{5}}}{4}, \quad \cos\phi_2 = -\frac{\sqrt{5}+1}{4}, \quad \sin\phi_2 = \frac{\sqrt{10-2\sqrt{5}}}{4}.$$

At the central pair the exact interval calculations in Appendix C give $P_j, H_j < -1/10000$ for all four displacements. Direct substitution into the physical residuals

$$p_i = (x_{i-1} - x_i)/r - y_{i-1}, \qquad q_i = y_{i-1} - y_i + (1+\theta)x_i - \theta x_{i-1}$$

identifies $\langle p_i, x_{i+j} - x_i\rangle = P_j$ and $\langle q_i, y_{i+j} - y_i\rangle = H_j$. Hence all forty nontrivial normal-cone inequalities hold, including the wraparound transition. The remaining equal-vertex tests are zero. Projection optimality gives

$$x_{i+1} = P_C(x_i - ry_i), \qquad y_{i+1} = P_D(y_i + (1+\theta)x_{i+1} - \theta x_i).$$

The first rotation plane has positive amplitude and primitive fifth order, so the five primal states are distinct.

For the entire rectangle, each primal test is affine in $1/r$ and independent of θ; each dual test is affine in θ and independent of r. The exact endpoint bounds in the same appendix remain below $-1/10000$. Thus the same normal inequalities, same states and same fixed hulls work at every point of (107).

Both hulls are full-dimensional simplices. Indeed, for any real coefficients α_i with $\sum_i \alpha_i = 0$, a relation $\sum_i \alpha_i x_i = 0$ forces the real and imaginary Fourier coefficients at both nonzero frequencies to vanish, because $A_1, A_2 > 0$. Together with the zero-frequency relation these five Fourier coefficients determine all α_i, so each is zero. The same argument applies to y_i, as $c_k^2 + d_k^2 > 0$ for both k. Also $\sum_i x_i = \sum_i y_i = 0$, so zero has strictly positive barycentric coordinates $1/5$ and lies in the interior of both hulls. It follows that $(0,0)$ is KKT. If (x,y) is any KKT pair, testing $-y \in N_C(x)$ and $x \in N_D(y)$ against zero gives $\langle x, y\rangle = 0$. If $y \neq 0$, a point $-\epsilon y \in C$ with $\epsilon > 0$ yields $\epsilon\|y\|^2 + \langle x, y\rangle \leq 0$, a contradiction. Thus $y = 0$, and $x \in N_D(0) = \{0\}$.

For the central parameters the comparisons are exact:

$$\frac{4}{1+2\theta} - r = \frac{1}{850}, \qquad (1+\theta)^2 r^2 - 2(1+3\theta)r - 3 = -\frac{124071}{6250000}. \tag{109}$$

The polynomial in r has positive leading coefficient and negative constant coefficient, hence exactly one positive root $R_3(\theta)$. Its negative value proves $0 < r < R_3(\theta)$. For the rectangle, $r(1+2\theta) < 4$ is checked at its upper right corner. The polynomial $Q(\theta, r) =$

$(1+\theta)^2 r^2 - 2(1+3\theta)r - 3$ is convex separately in each variable, so its value on a rectangle is bounded above by the largest corner value. All four exact corner values are negative. This proves (108) everywhere claimed. ∎

Consequently, the exact period-three onset and the bilinear period-two obstruction do not exhaust the possible nonconvergence mechanisms of the literal method in arbitrary Hilbert spaces. Their pointwise minimum cannot, by itself, be a universal sufficient convergence boundary for all positive extrapolations. The example is outside every sufficient interval proved here; it is compatible with the preceding sharp convergence theorems and with the stated ranges of the cited results. It gives neither the exact onset of five-cycles nor the full convergence boundary. The bounded literature comparison in Section 9 provides context for the certificate and periodic constructions; no exhaustive priority claim is made.

11 The five cycle onset and its transitions

The example in Theorem 7 lies inside a classified parameter mechanism. On a whole interval around 9/50, the first five-cycle is an isolated analytic algebraic branch. Its two crossings with the shorter-period thresholds are transverse, and its endpoint moments are forced to use both temporal frequencies. These statements resolve the onset and its changes of ordering, rather than adding another isolated feasible parameter pair.

The ingredients have important antecedents. The exact three- and four-cycle classifications are established in the earlier Reports 21 and 23 [7, 8]. Report 24 gives the complete two-frequency five-cycle conic criterion, including Gram slack; Report 25 gives the arbitrary-period criterion, realization, product-ray and frequency-support principles [9, 10]. Report 27 already uses the five-active-row discriminant and positive dual certificates to determine degree-twelve five-cycle onsets at 1/8 and 1 [11]. We use those methods here. The additional conclusions are a parameter-continuous onset classification, its exact competing-obstruction transitions, and normalized endpoint moment rigidity. The older statements have finite-dimensional scope; the finite-temporal Hilbert-space reduction below supplies the larger scope used here, without attributing it to those statements.

Put

$$I_5 = [177/1000, 183/1000], \qquad R_2(\theta) = \frac{4}{1+2\theta}, \qquad Q_3(\theta, r) = (1+\theta)^2 r^2 - 2(1+3\theta)r - 3. \tag{110}$$

The unique positive zero of Q_3 is the already defined R_3.

Theorem 8. *There is a real-analytic function ρ_5 on a neighborhood of I_5 with the following properties.*

1. *For $\theta \in I_5$ and $r > 0$, some fixed proper lower-semicontinuous convex problem for* (51)*, on real Hilbert spaces with bounded linear coupling and actual product r, has a nonconstant five-periodic orbit if and only if $r \geq \rho_5(\theta)$. The equivalence is unchanged when a KKT pair is required. At equality there is an identity-coupled realization by two fixed convex-hull indicators in $\mathbb{R}^4$. Every nonconstant five-periodic paired orbit has least period five.*

2. *There are exactly two comparison crossings in I_5:*

$$\rho_5(\theta_-) = R_3(\theta_-), \qquad \rho_5(\theta_+) = R_2(\theta_+), \qquad \theta_- < \theta_+.$$

   Both crossings are transverse, and

$$\rho_5(\theta) < \min\{R_2(\theta), R_3(\theta)\} \quad \Longleftrightarrow \quad \theta_- < \theta < \theta_+. \tag{111}$$

   Thus five-cycles strictly preempt every period-two, -three and -four obstruction precisely on this open interval.

3. *At $r = \rho_5(\theta)$, the feasible moments* (103)*, normalized by $A_1 + A_2 = 1$, are unique. Both frequencies are nonzero and both Gram cones are saturated.*

The algebraic branch and the two crossing numbers are defined exactly by (119)*,* (120) *and* (131) *below.*

For orientation only,

$$\begin{aligned} \theta_- &= 0.17800281318043712221\ldots, \\ \theta_+ &= 0.18160813025983395762\ldots, \\ \rho_5(9/50) &= 2.93901184220622775950\ldots. \end{aligned} \tag{112}$$

The theorem is classwise: the problem may depend on θ, r, but its two functions are fixed throughout the entire run. It neither classifies cycles of a prescribed problem nor asserts uniqueness of a physical embedding or a minimum spatial dimension. No claim is made that $\min\{R_2, R_3, \rho_5\}$ is a convergence boundary, or that these periods exhaust periodic or nonperiodic failure.

11.1 The full moment system on arbitrary Hilbert spaces

We first extend the necessity part of Lemma 3 to arbitrary bounded coupling. For a five-periodic orbit, all values of f and h at its phases are finite because every phase

is a proximal output. Sum the subgradient inequalities from phase i to $i+j$, for each $j = 1, \dots, 4$. The function values telescope. Center the tuples and set

$$U_i = (x_i - \bar{x})/\sqrt{\tau}, \quad V_i = (y_i - \bar{y})/\sqrt{\sigma}, \quad K_0 = \sqrt{\tau\sigma}L, \quad r = \|K_0\|^2.$$

Equip the temporal direct sums with the phase-average inner product, write T for forward shift and $S = T^{-1}$ for backward shift, and use the same notation for their separate primal and dual versions. The necessary inequalities are

$$\begin{aligned} \langle (S-I)U - K_0^* SV, (T^j - I)U \rangle &\le 0, \\ \langle (S-I)V + K_0((1+\theta)I - \theta S)U, (T^j - I)V \rangle &\le 0. \end{aligned} \tag{113}$$

The constant terms removed by centering are orthogonal to the cyclic displacements. The lift of K_0 intertwines the two shifts.

The centered five-phase space is the orthogonal sum of two real frequency subspaces. Its projections are finite polynomials in T, T^*, so this decomposition is valid without finite spatial dimension or separability. On frequency k, $T = C_k I + \sin\phi_k J_k$, where $J_k^* = -J_k$, $J_k^2 = -I$, $\phi_k = 2\pi k/5$, and $C_k = \cos\phi_k$. Set

$$A_k = r\|U_k\|^2, \quad B_k = \|V_k\|^2, \quad c_k = \langle K_0 U_k, V_k \rangle, \quad d_k = \langle K_0 J_k U_k, V_k \rangle. \tag{114}$$

The vectors $K_0 U_k$ and $K_0 J_k U_k = J_k K_0 U_k$ are orthogonal and have the same norm. Bessel's inequality and the actual norm bound therefore give

$$c_k^2 + d_k^2 \le \|K_0 U_k\|^2 B_k \le A_k B_k.$$

If $K_0 U_k = 0$, the correlations vanish and the same inequality holds. Expanding (113) gives exactly the eight rows (102), including every sign and the factor A_k/r. If $U = 0$, the last dual displacement forces $\|(S-I)V\|^2 \le 0$, hence $V = 0$ after centering. Thus a nonconstant orbit with $r > 0$ has $A_1 + A_2 > 0$, and common positive rescaling gives the stated normalization. No norm maximizer, injectivity or finite-rank assumption has entered. If $r = 0$, then $L = 0$ and the two proximal-point sequences are constant: summing their predecessor subgradient inequalities forces the sums of squared increments to vanish.

Conversely, Lemma 3 already realizes the entire conic system, retaining singular modes and Gram slack. Its finite-dimensional construction belongs to the present larger class. The moment criterion is therefore an equivalence here as well.

11.2 An algebraic active face and its analytic continuation

Use $s = \sqrt{5}$ and $h_1 = (5+s)/8$, $h_2 = (5-s)/8$, and change coordinates to

$$w = (X_1, X_2, B_1, B_2, c_1, c_2, e_1, e_2)^T, \qquad A_k = rX_k, \quad d_k = \sqrt{h_k}\, e_k. \tag{115}$$

All eight interpolation rows are now independent of r, and the cones are $X_k, B_k \geq 0$ and

$$q_k(w) = rX_kB_k - c_k^2 - h_ke_k^2 \geq 0. \tag{116}$$

Let $M(\theta)$ be the five-row matrix formed, in order, from P_1, P_3, H_1, H_2, H_3 in these coordinates. A basis for its nullspace is the matrix

$$N(\theta) = \begin{pmatrix} 0 & (-3\theta + s\theta - 4)/4 & \theta/2 \\ 0 & \theta/2 & -(s\theta + 3\theta + 4)/4 \\ \theta + 1 & (-5\theta^2 + s\theta^2 - 5\theta + s\theta - 2)/2 & (s+5)(\theta+1)(2\theta+1)/4 \\ \theta + 1 & 0 & (s\theta + 5\theta + 1 + s)/4 \\ 1 & (-5+s)(2\theta+1)/4 & (s+5)(2\theta+1)/4 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{117}$$

Indeed $MN = 0$, and the determinant of the first five columns of M is the nonzero constant $125\sqrt{5}/16$. Thus this is the entire nullspace for every θ, with free coordinates (c_2, e_1, e_2). Specializing N at $1/8$ or 1 recovers the active-face matrices of Report 27; the method itself is inherited.

For a row N_a of N, define the symmetric 3×3 forms

$$Q_k = \frac{r}{2}(N_{X_k}^T N_{B_k} + N_{B_k}^T N_{X_k}) - N_{c_k}^T N_{c_k} - h_k N_{e_k}^T N_{e_k}, \qquad D(\theta, r, z) = \det(Q_1 + zQ_2). \tag{118}$$

Write $D = az^3 + bz^2 + cz + d$. A compact exact defining polynomial is

$$\mathcal{F}(\theta, r) \doteq \operatorname{disc}_z D = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd, \tag{119}$$

where $\doteq$ permits one fixed nonzero scalar factor. The rational normalization in the ancillary exact data has bidegree $(20, 12)$ and 195 nonzero terms. This determinant definition, rather than a list of large coefficients, specifies its mathematical origin. The replay reconstructs it exactly from the eight rows.

Here is a finite rational specification of the selected root branch. For $j = 0, \ldots, 1023$, set

$$\begin{aligned} I_j &= \left[\frac{177}{1000} + \frac{3j}{512000}, \frac{177}{1000} + \frac{3(j+1)}{512000}\right], & t_j &= \operatorname{mid}(I_j), \\ \delta_j &= t_j - \frac{9}{50}, & \eta &= \frac{3}{256000} + \frac{1}{5000000}, \\ v_j &= \frac{2939011842206228}{10^{15}} - \frac{29748030786}{10^{10}}\delta_j + \frac{36136664238}{10^{10}}\delta_j^2, & J_j &= [v_j - \eta, v_j + \eta]. \end{aligned} \tag{120}$$

In this normalization the exact rational certificate proves, uniformly,

$$\mathcal{F}(\theta, v_j - \eta) < 0 < \mathcal{F}(\theta, v_j + \eta) \quad (\theta \in I_j), \qquad \mathcal{F}_r > 0 \quad \text{on } I_j \times J_j. \tag{121}$$

Thus for each $\theta \in I_j$ there is exactly one zero in J_j. The decimal-looking rational coefficients in v_j merely locate convenient boxes; no fitted value is used as a proof premise. Adjacent J_j overlap. At their shared θ endpoint, $\mathcal{F}_r > 0$ throughout the connected union of the two root intervals. Consequently the two zeros coincide. This extra monotonicity argument is necessary: interval overlap alone would not identify roots. The exact base isolator at $9/50$ lies strictly inside the corresponding J_j. These facts glue one graph $r = \rho_5(\theta)$. The implicit-function theorem and the strict bounds on a compact cover extend the graph analytically to a neighborhood of I_5. Appendix D gives the replay and the precise interval arithmetic used to establish these continuous signs.

On the graph define

$$z = \frac{9ad - bc}{2(b^2 - 3ac)}, \quad K = Q_1 + zQ_2, \quad u = \frac{-K_{1,:} \times K_{2,:}}{(K_{1,:} \times K_{2,:})_3}, \quad w = Nu. \tag{122}$$

The same continuous certificate proves $2(b^2 - 3ac) > 0$, $z > 0$, $D_{zz} \neq 0$, a nonzero denominator in u, $X_k, B_k > 0$, and

$$P_1 = P_3 = H_1 = H_2 = H_3 = 0, \qquad P_2, P_4, H_4 < 0. \tag{123}$$

The cubic discriminant identity gives $D = D_z = 0$ at this z. The nonzero cross product and zero determinant make K rank two, with kernel vector u and $u_3 = -1$. The two cone equalities follow exactly, not from small numerical residuals. Namely $u^T K u = u^T Q_1 u + z u^T Q_2 u = 0$. For a symmetric rank-two matrix its adjugate is a nonzero scalar multiple of uu^T. Jacobi's formula and $D_z = 0$ therefore give $u^T Q_2 u = 0$, and then $u^T Q_1 u = 0$. These are $q_1(w) = q_2(w) = 0$. Dividing all coordinates of w by $\rho_5(X_1 + X_2)$ gives $A_1 + A_2 = 1$.

11.3 A positive dual proves the global onset

It remains essential to show that this branch is the first feasible product over the full moment system, not just on the chosen active face. Fix $\theta \in I_5$, put $\rho = \rho_5(\theta)$, and use the unnormalized witness in (122). Let $\mu_1 = 1$, $\mu_2 = z$ and set $g = \nabla_w(q_1 + zq_2)$ at that witness:

$$g_{X_k} = \mu_k \rho B_k, \quad g_{B_k} = \mu_k \rho X_k, \quad g_{c_k} = -2\mu_k c_k, \quad g_{e_k} = -2\mu_k h_k e_k. \tag{124}$$

For the first five columns $J = (1, 2, 3, 4, 5)$ define $\lambda = M_J^{-T} g_J$. Exact interval bounds give every $\lambda_i > 0$ throughout I_5. Moreover

$$M^T \lambda = g. \tag{125}$$

Indeed $N^T g = 2Ku = 0$, so g lies in the row space of the rank-five matrix M. Solving its five pivot coordinates establishes all eight coordinates, not merely a partial system.

To test an arbitrary feasible vector at a possibly smaller product r, distinguish it by tildes and return to its physical moment coordinates $\widetilde{A}_k = r\widetilde{X}_k$, $\widetilde{d}_k = \sqrt{h_k}\widetilde{e}_k$. Its weighted interpolation sum is, by (125),

$$\sum_{k=1}^{2} \mu_k \left\{ \frac{\rho}{r} B_k \widetilde{A}_k + \rho X_k \widetilde{B}_k - 2c_k \widetilde{c}_k - 2d_k \widetilde{d}_k \right\}. \tag{126}$$

At $r = \rho$, each bracket is a positive semidefinite rank-one functional on the Hermitian Gram cone $\widetilde{A}_k \widetilde{B}_k \geq \widetilde{c}_k^2 + \widetilde{d}_k^2$. Its two diagonal coefficients are positive and its determinant is zero because $c_k^2 + d_k^2 = \rho X_k B_k$. For $0 < r < \rho$, the determinant of the representing 2×2 Hermitian form is

$$\mu_k^2 B_k \rho X_k \left(\frac{\rho}{r} - 1 \right) > 0. \tag{127}$$

Equivalently its Lorentz discriminant is four times this quantity. Thus each mode functional is strictly positive on every nonzero positive semidefinite Gram matrix. This includes a zero diagonal coordinate and arbitrary Gram slack. A nonconstant orbit has $\sum \widetilde{A}_k > 0$, so (126) is strictly positive. But every active interpolation row is nonpositive and its multiplier is positive, a contradiction. This excludes every smaller product universally. The hypothetical feasible point was never restricted to the active face, and no monotonicity of individual signed D_{kj} coefficients was used.

At the endpoint the witness satisfies all eight inequalities and both cones. For every $r \geq \rho$, keep $w = (X, B, c, e)$ fixed. The rows do not change, and each cone increases by $(r - \rho) X_k B_k \geq 0$. This proves feasibility on the entire upper ray. If the endpoint data were normalized, multiply this fixed w by ρ/r when normalizing again: the new $A = rX$ must have sum one. This is the normalized upper-ray rescaling already established in Report 25.

For completeness, the endpoint realization is literal. With the normalized saturated moments take two orthogonal rotation planes and

$$x_{0,k} = (\sqrt{A_k}, 0), \qquad y_{0,k} = (c_k/\sqrt{A_k}, d_k/\sqrt{A_k}), \qquad x_i = \mathcal{R}^i x_0, \quad y_i = \mathcal{R}^i y_0. \tag{128}$$

Here $\mathcal{R} = \operatorname{diag}(\mathcal{R}(\phi_1), \mathcal{R}(\phi_2))$. Let $C = \operatorname{conv}\{x_i\}$, $E = \operatorname{conv}\{y_i\}$, $f = \iota_C$, $h = \iota_E$, $L = I$, $\tau = \rho$, $\sigma = 1$. Direct rotation multiplication gives, for all i and $j = 1, \ldots, 4$,

$$\begin{aligned} \left\langle \frac{x_{i-1} - x_i}{\rho} - y_{i-1}, x_{i+j} - x_i \right\rangle &= P_j \leq 0, \\ \langle y_{i-1} - y_i + (1+\theta) x_i - \theta x_{i-1}, y_{i+j} - y_i \rangle &= H_j \leq 0. \end{aligned} \tag{129}$$

These forty vertex comparisons, together with the zero diagonal ones, are exactly the two hull normal-cone conditions. The first input-minus-output residual is ρ times its displayed normal; the second is its displayed normal. Unique projections therefore give the literal next paired states, from the ordinary initialization (x_0, y_0) and with no hidden prehistory. Both primitive frequencies are nonzero, so the primal states are distinct and the least period is five. Since $\sum_{i=0}^{4} \mathcal{R}^i = 0$, the hulls contain zero and $(0, 0)$ is KKT.

Above onset, an extra plane in each mode accommodates $\sqrt{B_k - (c_k^2 + d_k^2)/A_k}$ as in the earlier realization proof, using at most $\mathbb{R}^8$. The same normal identities retain that entire slack. The zero-A_k convention is $x_k = 0$ and a dual vector of norm $\sqrt{B_k}$, with no division by zero. This completes the existence and exclusion part of Theorem 8.

11.4 Endpoint moment rigidity

At $r = \rho$, the positive weighted interpolation sum is nonpositive by feasibility and nonnegative by its two cone functionals. Hence both cone functionals vanish and all five positively weighted rows vanish. A nonzero positive semidefinite rank-one functional on the 2×2 Hermitian positive cone exposes a single nonnegative ray: if $H \geq 0$ and $\operatorname{tr}(BH) = 0$ for rank-one $B \geq 0$, then the range of H is contained in the one-dimensional kernel of B. Thus every feasible endpoint mode is a nonnegative multiple β_k of the reference mode.

The remaining possibility of independent mode scaling is excluded by an interpolation equality. Write $P_1^{(k)}$ for the contribution of mode k to P_1. The continuous rigidity certificate proves

$$P_1^{(1)} < 0 \quad \text{throughout } I_5. \tag{130}$$

Since the reference vector has $P_1 = 0$, it has $P_1^{(2)} = -P_1^{(1)} > 0$. For a feasible endpoint vector, $0 = \beta_1 P_1^{(1)} + \beta_2 P_1^{(2)}$ forces $\beta_1 = \beta_2$. Normalizing $A_1 + A_2 = 1$ fixes the common positive multiple. Both frequencies and both saturated cones are therefore necessary for every feasible normalized endpoint moment vector. This proves rigidity for the full conic system, rather than merely for its manufactured active face. It does not identify all spatial orbits or all convex functions realizing the same moments.

11.5 The two exact transverse crossings

Let pp mean the primitive polynomial numerator after clearing rational denominators and removing a nonzero scalar. Define

$$\begin{aligned} G_2(\theta) &= \operatorname{pp} \mathcal{F}\left(\theta, \frac{4}{1+2\theta}\right), \\ G_3(\theta) &= \operatorname{pp}\left[\frac{\operatorname{Res}_r(\mathcal{F}(\theta, r), Q_3(\theta, r))}{(1+\theta)^8}\right]. \end{aligned} \tag{131}$$

Their degrees are 20 and 32. Exact Sturm counts show that each has exactly one real root in I_5, and exact polynomial gcd calculations show both are squarefree. Define θ_+ as the root of G_2 and θ_- as the root of G_3 there. The rational isolators in the ancillary certificate are disjoint and satisfy

$$\theta_- < \theta_0 := \frac{4\sqrt{5} - 5}{22} < \theta_+. \tag{132}$$

At $\theta = 177/1000$, exact endpoint signs give $\rho_5 - R_2 < 0$ and $Q_3(\theta, \rho_5) > 0$; at $183/1000$ both signs reverse. Because $\rho_5 > 0$ and Q_3 has exactly one positive root, its sign at ρ_5 is the sign of $\rho_5 - R_3$. Continuity forces a genuine crossing of each comparison. Every genuine crossing is a root of its corresponding polynomial. Sturm uniqueness therefore excludes an extraneous negative-R_3 root or a different $\mathcal{F}$ branch as the sole explanation for the resultant root. It also excludes further crossings. We obtain

$$\begin{array}{ll} \rho_5 > R_3 \text{ left of } \theta_-, & \rho_5 < R_3 \text{ right of } \theta_-, \\ \rho_5 < R_2 \text{ left of } \theta_+, & \rho_5 > R_2 \text{ right of } \theta_+. \end{array} \tag{133}$$

Squarefreeness proves more than a change of sign. The zero of $\mathcal{F}(\theta, R_2(\theta))$ at θ_+ is simple because the cleared denominator is nonzero on I_5. Since $\mathcal{F}_r \neq 0$ on the selected branch, $\rho_5 - R_2$ has a simple zero there. For R_3, the two roots of Q_3 are distinct and analytic locally, and its leading coefficient is nonzero. The local product formula for the resultant expresses it as a nonvanishing factor times $\mathcal{F}(\theta, R_3(\theta))\mathcal{F}(\theta, R_3^-(\theta))$. Its simple zero forces the first factor to have a simple zero and the second to be nonzero. Again $\mathcal{F}_r \neq 0$ gives a simple zero of $\rho_5 - R_3$. Both crossings are therefore transverse.

The exact identity

$$Q_3(\theta, R_2(\theta)) = -\frac{44\theta^2 + 20\theta - 5}{(1 + 2\theta)^2} \tag{134}$$

locates the already known two/three crossing at θ_0. Together the three numbers give the complete strict ordering

$$\begin{array}{c|l} [177/1000, \theta_-) & R_3 < \rho_5 < R_2 \\ (\theta_-, \theta_0) & \rho_5 < R_3 < R_2 \\ (\theta_0, \theta_+) & \rho_5 < R_2 < R_3 \\ (\theta_+, 183/1000] & R_2 < \rho_5 < R_3. \end{array} \tag{135}$$

At the three omitted boundaries exactly the named equalities hold. In particular $9/50 \in (\theta_0, \theta_+)$ and $\rho_5(9/50) < 147/50 < R_2(9/50) < R_3(9/50)$. The explicit fixed-hull rectangle of Theorem 7 is therefore a supercritical example inside this classified phase. Its original proof and stronger fixed-hull persistence statement remain useful; the onset theorem does not require one pair of hulls to work for different parameters.

11.6 Why period four cannot preempt this phase

We record the short-period necessity arguments to keep the comparison valid in the present Hilbert-space scope. In the period-two moment system, normalize its one nonconstant primal mode to $A = 1$ and write $\alpha = 1 + 2\theta > 0$. Its two cyclic tests and Gram bound give

$$c \geq 2/r, \qquad B \leq \alpha c/2, \qquad c^2 \leq B.$$

Consequently $c > 0$, $c \leq \alpha/2$ and $r \geq 4/\alpha = R_2$. At equality $c = 2/r$, $B = c^2$ gives a fixed-indicator segment realization with KKT, by the same normal-cone calculation. These are

the period-two conclusions of Report 25; the norm/Bessel reduction just proved is valid in arbitrary real Hilbert spaces. The period-three onset and attainment in this scope have already been proved in Theorems 3 and 4.

For period four, use centered and scaled U, V, K_0 as above, now on fourfold temporal sums with backward shift S. Split the centered space into its quarter-turn mode $S^2 = -I$ and its alternating mode $S = -I$, with components U_1, V_1 and U_2, V_2. These are bounded orthogonal projections given by $(I - S^2)/2$ and $(I + S^2)/2$ on centered tuples. Set

$$A = \|U_1\|^2, \quad B = \|V_1\|^2, \quad E = \|U_2\|^2, \quad F = \|V_2\|^2,$$
$$c = \langle V_1, K_0 U_1 \rangle, \quad q = \langle V_1, K_0 S U_1 \rangle, \quad g = \langle V_2, K_0 U_2 \rangle.$$

The displacement-one and half displacement-two tests are

$$P = 4E + c - q - 2g \le 0, \quad H = 4F - c + \alpha q - 2\alpha g \le 0,$$
$$P' = A - q \le 0, \qquad\qquad H' = B - (1+\theta)c + \theta q \le 0.$$

The strict-bound identity from Report 23 is

$$\begin{aligned} 0 &\ge \alpha P + H + \frac{2\theta}{1+\theta} H' + \frac{2\theta^2}{1+\theta} P' \\ &= 4 \left\| V_2 - \frac{\alpha}{2} K_0 U_2 \right\|^2 + \alpha(4 - \alpha r)E + \alpha^2 (rE - \|K_0 U_2\|^2) \\ &\quad + \frac{2\theta}{1+\theta} B + \frac{2\theta^2}{1+\theta} A. \end{aligned} \tag{136}$$

If $r \le R_2$ and $\theta > 0$, every term is nonnegative and the last two coefficients are positive. Thus $A = B = 0$, leaving only the alternating mode and constants: the paired period divides two. Every least-period-four orbit consequently has $r > R_2$. This finite-temporal proof uses only the norm bound and remains valid on arbitrary real Hilbert spaces. Combining it with (133) proves the claimed exact strict preemption interval among periods two through five, including the ties at its endpoints. It does not turn this finite-period comparison into a convergence theorem.

A Exact data and reproducible certificate

This appendix specifies the entire mathematical meaning of the supplied coefficient file and the finite checks used in Lemma 1. The JSON file supplies the exact numbers, not numerical approximations. A previously generated results file is never trusted as an input to either verifier.

A.1 Field arithmetic and data schema

The field is $\mathbb{F} = \mathbb{Q}(\sqrt{91})$. A serialized pair $[a, b]$ denotes $a + b\sqrt{91}$; a, b are strings encoding rational numbers. Addition and multiplication are

$$(a, b) + (c, d) = (a + c, b + d), \qquad (a, b)(c, d) = (ac + 91bd, ad + bc).$$

For a nonzero divisor, inversion uses $(a+b\sqrt{91})^{-1} = (a-b\sqrt{91})/(a^2-91b^2)$. When a and b have opposite signs, the sign of $a+b\sqrt{91}$ is resolved by the exact rational comparison of a^2 and $91b^2$. They cannot be equal unless $a=b=0$, since 91 is not a rational square. No decimal conversion enters a proof check.

The file `exact_certificate.json` has the following schema.

- `description`: human-readable provenance text, with no mathematical role in the checks.
- `rstar`: the pair $[88/81, 16/81]$; `delta`: the rational string 1/100000; `m`: the integer 4.
- `P_numerator`: three arrays, each of length 36, giving the upper triangular entries of P_0, P_1, P_2. The ordering is lexicographic in (i,j) with $0 \le i \le j < 8$; reflection supplies the lower triangle.
- `c`: two arrays of length 8, giving c_0, c_1.
- `lambda`: two arrays of length 50, giving λ_0, λ_1 in the edge order below.
- `mu`: two arrays of length 32, giving μ_0, μ_1 in the inherited lower-bound edge order.

Each scalar entry in the last four fields is a field pair. Thus the file contains 288 field entries in these coefficient arrays: $108 + 16 + 100 + 64$. Only P is divided by r in (23); the other three families are not multiplied or divided by r.

For a small directly readable part of the data, $c_0 = 0$ and

$$c_1 = \Bigg(0, 0, -\frac{16896789}{2500000000}, -\frac{209}{5000000000}, \\ -\frac{33793453}{5000000000}, \frac{3}{1250000000}, -\frac{2703483}{400000000}, \frac{197}{5000000000}\Bigg)^T. \tag{137}$$

This illustrates why positivity of the value part alone cannot be used. The complete matrix and multiplier arrays are in the accompanying exact file, identified by the SHA-256 digest

```
421ea0abb9e4bb550d19310aa0c85d90
db54531b181c1378796904bff5b97815
```

(the two lines are concatenated, without whitespace, to give the digest).

A.2 Slot and edge conventions

All vector and value slots have the order $(X_0, Y_0, X_1, Y_1, \ldots, X_4, Y_4)$, indexed from 0 through 9. Let $e_{i,s} \in \mathbb{R}^{10}$ be the selector for position $2i+s$, with $s=0$ primal and $s=1$

dual. Use an anchor symbol $*$ whose selector, point, value, and subgradient are zero. For $i \geq 1$ the gradient selectors are

$$\begin{aligned} g_{i,0} &= (e_{i-1,0} - e_{i,0})/r - e_{i-1,1}, \\ g_{i,1} &= e_{i-1,1} - e_{i,1} + (9e_{i,0} - e_{i-1,0})/8. \end{aligned} \tag{138}$$

There is no subgradient at formal index 0. For an edge (i, j, s), define $d_{i,j,s} = e_{j,s} - e_{i,s}$ with the anchor convention, and set

$$v_{i,j,s} = d_{i,j,s}, \qquad Q_{i,j,s}(r) = -\operatorname{sym}(g_{i,s} d_{i,j,s}^T), \quad \operatorname{sym}(A) = (A + A^T)/2. \tag{139}$$

Then (19) is precisely $\phi_{j,s} - \phi_{i,s} - \langle g_{i,s}(w), z_{j,s} - z_{i,s} \rangle$. For $i = *$ the quadratic matrix is zero. The edges $(*, j, s)$ are the nonnegative anchored values; edges $(i, *, s)$ are the upper anchor inequalities.

For each $s = 0, 1$, in that order, the list $\mathcal{A}$ is

1. $(*, j, s)$ for $j = 0, 1, 2, 3, 4$;
2. $(i, *, s)$ for $i = 1, 2, 3, 4$;
3. (i, j, s) for $i = 1, 2, 3, 4$ and $j = 0, 1, 2, 3, 4$, $j \neq i$, lexicographically in (i, j).

There are $5 + 4 + 16 = 25$ edges per variable. Delete all edges with $i = 4$ or $j = 4$, retaining order, to obtain $\mathcal{B}$ with 16 per variable. Thus $\mathcal{B}$ uses only the stored four-pair history. The selectors are $O = [I_8 \ 0_{8\times 2}]$ and $N = [0_{8\times 2} \ I_8]$. These definitions determine every matrix and value vector in (20)–(21).

A.3 Reconstructing the polynomial identities

Write $Q_e(r) = Q_e^{\mathrm{diff}}/r + Q_e^{\mathrm{cross}}$, where for a primal edge

$$Q_e^{\mathrm{diff}} = -\operatorname{sym}((e_{i-1,0} - e_{i,0}) d_e^T), \qquad Q_e^{\mathrm{cross}} = \operatorname{sym}(e_{i-1,1} d_e^T),$$

and for a dual edge $Q_e^{\mathrm{diff}} = 0$ and $Q_e^{\mathrm{cross}} = -\operatorname{sym}(g_{i,1} d_e^T)$. Both parts vanish for an anchor-source edge. Then

$$r Q_e(r) = Q_e^{\mathrm{diff}} + r_\star Q_e^{\mathrm{cross}} - t Q_e^{\mathrm{cross}} = Q_{e,0} + t Q_{e,1}.$$

The three coefficients of rD and rC are obtained by ordinary polynomial multiplication: the numerator copies of P are placed using O, N, and each affine multiplier is multiplied by $Q_{e,0} + t Q_{e,1}$. The two separate value identities are affine and are checked at the coefficient level, not by evaluation at a few parameter values. For every multiplier $a + tb$, the exact tests are $a \geq 0$ and $a + \delta b \geq 0$.

The verifiers also check, for each of the three coefficient matrices, that $rDZ = 0$, that all entries of rC touching slots 8 or 9 are zero, and that the two diagonal identities in (22) hold after multiplying by r. At $t = 0$, they check $P_0 Z_8 = 0$ (or the corresponding zero constant-history form). These are algebraic identities of the provided data and the raw proximal slacks.

A.4 Explicit kernel bases and interval checks

Here are the concrete bases used by the first verifier. Define the two constant columns Z, the two ramp columns R_0, and two cycle columns A_0 by their pair rows, for $i = 0, 1, 2, 3, 4$:

$$Z_{(i)} = I_2, \qquad (R_0)_{(i)} = iI_2,$$

$$(A_0)_{(i)} = \begin{pmatrix} q_i & s_i \\ \frac{9}{16}q_i + \frac{3}{2r_\star}s_i & \frac{9}{16}s_i - \frac{1}{2r_\star}q_i \end{pmatrix}, \quad (q_i, s_i) = \begin{cases} (1,0), & i \equiv 0 \pmod 3, \\ (-1/2, 1/2), & i \equiv 1 \pmod 3, \\ (-1/2, -1/2), & i \equiv 2 \pmod 3. \end{cases} \tag{140}$$

The full endpoint descent numerator annihilates $[Z, R_0, A_0]$. Its exact rank is four; these six columns span its kernel. The endpoint coercivity numerator has leading-block rank six and annihilates Z_8.

For the descent leading block, set $W = [R_0, A_0] \in \mathbb{R}^{10\times 4}$ and

$$K_D = W_{0:8,:} - Z_8 W_{8:10,:}, \qquad E_D = [e_0, e_1, e_2, e_3] \in \mathbb{R}^{8\times 4}. \tag{141}$$

The notation $0:8$ means rows 0 through 7. For the coercivity block use

$$K_C = Z_8, \qquad E_C = [e_0, e_1, \ldots, e_5] \in \mathbb{R}^{8\times 6}. \tag{142}$$

In both cases $T = [E, K]$ is exactly invertible and $Q_0 K = 0$. The second verifier independently computes the endpoint nullspaces by rational-field row reduction, rather than importing these basis matrices. It verifies ranks four and six, constructs a complement from the first four or six coordinate columns, and separately reconstructs the full constant/ramp/mean-zero period-three kernel geometry.

For either basis, the block coefficients in (24) are $A_j = E^T Q_j E$, $B_j = E^T Q_j K$, $C_j = K^T Q_j K$; the zero constant cross and kernel blocks are verified exactly. Construct (25) and its three controls (26). For a symmetric matrix B, the unpivoted LDL^T calculation uses

$$d_j = B_{jj} - \sum_{k<j} \ell_{jk}^2 d_k, \qquad \ell_{ij} = \frac{B_{ij} - \sum_{k<j} \ell_{ik}\ell_{jk}d_k}{d_j} \quad (i > j), \qquad \ell_{jj} = 1.$$

All eight pivots must be strictly positive. The reconstructed matrix $L\,\mathrm{diag}(d)L^T$ must equal B exactly. The six checks produce 48 exact positive pivots per implementation. The result files store them as rational-field pairs. The second implementation additionally checks that the controls recover H_0, H_1, H_2 and that the congruence (28) holds coefficientwise through degree four. These finite tests, together with the displayed interval proof, certify every $0 < t \le \delta$, with no sampling.

A.5 Provenance and reproduction

The coefficient file is a fixed exact algebraic witness. Its construction uses a semi-coercive endpoint storage and an affine deformation of free coordinates, which produces a quadratic numerator for P. The mathematical provenance needed for reproducibility is the complete rational-field data, the raw proximal-slack reconstruction, and the exact checks above. An optimization solver, a saved numerical state, or a discovery program is not part of the proof's trusted input. In particular, reproducing a numerical search is unnecessary for validating the result.

The source archive contains the manuscript and its unchanged style file. It also supplies the exact coefficient file, a human-readable witness specification, the theorem statement, two Python verifiers, and their regenerated exact results. The first verifier directly implements the explicit kernel bases above. The second independently implements field signs, proximal slack construction, row reduction, interval congruences, and an endpoint normal cone check; it imports no code or output from the first verifier. Both use only the Python standard library. Their public paths are relative to the source directory. A hash manifest identifies their inputs.

From the extracted source directory, run

```
python3 verify.py
python3 verify_independent.py
```

Both commands must complete with `PASS`. They overwrite their own result files after recomputing all checks, including the endpoint projection inequalities. Assertions must remain enabled; do not run Python with its optimization flag. The archive includes a build script for two LaTeX passes and a hash-check script. A standard pdfLaTeX installation with the packages loaded by `reference-style.sty` is sufficient to rebuild the paper. The exact algebraic checks do not need LaTeX.

The paper and archive make a reproducible finite certificate available; the analytic steps connecting it to the original arbitrary-dimensional iteration are proved in Sections 2–4. This division is essential: positive matrix checks alone are not a substitute for the signed-value decay, arbitrary-initialization treatment, or KKT-restricted separation argument.

B The continuous rational certificate and its exact replay

This appendix proves Lemma 2 and specifies the public inputs completely. It supplements, without replacing, the fixed-1/8 data in Appendix A. The numerical size of the rational coefficients makes a machine-readable exact file preferable to pages of deci-

mal approximations; the schema and reconstruction below state what each entry means mathematically.

B.1 Rational functions and recovery of the fixed parameter data

The file `theta_certificate/family.json` contains metadata for R, θ, δ in (56) and a dictionary of 189 explicit rational functions in the single symbol u. Every expression uses integer literals, rational arithmetic and integer powers. Coefficient keys `P`k`_`j, for $k = 0, 1, 2$, give the 36 upper triangular entries of P_k in row-major order

$$(0,0), (0,1), \ldots, (0,7), (1,1), \ldots, (7,7).$$

Symmetry supplies the lower triangle. Keys `c`k`_`j use $k = 0, 1$, $j = 0, \ldots, 7$. Multiplier keys `l`k`_`j use $j = 0, \ldots, 49$, and keys `m`k`_`j use $j = 0, \ldots, 31$. All P, c entries are supplied; unlisted multiplier entries are exactly zero. Thus the 189 stored entries expand to $3 \cdot 36 + 2 \cdot 8 + 2 \cdot 50 + 2 \cdot 32 = 288$ scalar slots.

The old input `exact_certificate.json` remains unchanged. Its byte-identical local copy `theta_certificate/eighth_certificate.json` allows this supplement to replay independently of the parent directory. Each old coefficient is the exact pair $[a, b]$ for $a + b\sqrt{91}$. At

$$u_0 = \frac{11 + 2\sqrt{91}}{9}, \qquad 27u_0^2 - 66u_0 - 81 = 0, \tag{143}$$

substitution recovers all 288 slots, not merely P_0 or the endpoint. The checks reduce numerator polynomials modulo the displayed minimal polynomial and verify every denominator is nonzero. They also verify the old active multiplier support. The polynomial in (143) is strictly increasing on J, with values $-124413/1000000$ and $7563/160000$ at its respective endpoints. Thus u_0 lies strictly inside J. Exact substitution gives $\theta(u_0) = 1/8$ and $R(u_0) = r_\star$.

All original combined denominators of the 189 rational functions are checked before any residual cancellation; there are 60 distinct classes. This prevents cancellation from concealing an undefined input coefficient. The parameter checks also prove $R - \delta > 0$, $\theta > 0$, $\theta' > 0$, and the two identities following (56) throughout J.

B.2 Edges and literal reconstruction

The point and value slots are interleaved in the respective orders

$$(X_0, Y_0, \ldots, X_4, Y_4), \qquad (F_0, H_0, \ldots, F_4, H_4).$$

For each variable, an ordered edge (i, j) represents

$$S_{ij} = \phi_j - \phi_i - \langle g_i, z_j - z_i \rangle,$$

where the anchor $*$ has zero point, value and gradient. The residuals g_i for $i = 1, \dots, 4$ are exactly (59). For λ place all primal edges before all dual edges. Within each variable their order is

$$\begin{gathered}(*, j)\ (j = 0, \dots, 4), \qquad (i, *)\ (i = 1, \dots, 4), \\ (i, j)\ (i = 1, \dots, 4;\ j = 0, \dots, 4,\ j \neq i),\end{gathered} \tag{144}$$

with i the outer index in the last group. There are 25 edges per variable. For μ, remove every edge touching position 4, retaining the order, to obtain 16 per variable. No selected gradient at position 0 is ever used.

Writing a scalar slack as $w^T Q_e w + d_e^T \phi$ determines the symmetric matrix Q_e uniquely, with half the mixed coefficient in each symmetric entry. The independent literal verifier constructs it as half the Hessian of the literal scalar polynomial. Reconstructing (60)–(61) then verifies, coefficient by coefficient, that rD, rC are quadratic in t, that both value-flow identities vanish, that every coefficient of rD annihilates Z, and that every coefficient of rC has zero final two rows and columns. The two diagonal entries of $Z_8^T(P - C_8)Z_8$ vanish identically. Those algebraic identities imply (62) once $C_8 > 0$ is established below.

B.3 Endpoint kernels and an explicit nonsingular basis

Let the pair block of Ramp at position i be iI_2. Cycle (q_i, s_i) through

$$(1, 0), \qquad (-1/2, 1/2), \qquad (-1/2, -1/2),$$

and let A_0^{cyc} have pair rows

$$(q_i, s_i), \qquad \left(\frac{1+\theta}{2} q_i + \frac{3 s_i}{2R},\ \frac{1+\theta}{2} s_i - \frac{q_i}{2R}\right). \tag{145}$$

At $t = 0$, rD annihilates $[Z, \mathrm{Ramp}, A_0^{\mathrm{cyc}}]$; P_0 and rC_8 annihilate Z_8. These are exact polynomial identities. Write $W = [\mathrm{Ramp}, A_0^{\mathrm{cyc}}]$, and use

$$\begin{aligned} K_D &= W_{0:8} - Z_8 W_{8:10}, & E_D &= [e_0, e_1, e_2, e_3], \\ K_C &= Z_8, & E_C &= [e_0, e_1, \dots, e_5]. \end{aligned} \tag{146}$$

Row slices have zero-based, upper-exclusive endpoints. Direct exact calculation gives

$$\det[E_D, K_D] = \frac{18}{u^2 - 6u - 3}, \qquad \det[E_C, K_C] = 1. \tag{147}$$

The first denominator is $-4R < 0$, so both bases are nonsingular on J.

For either $Q(t) = rD_8$ or $Q(t) = rC_8$, use the corresponding $B = [E, K]$. Endpoint annihilation gives

$$B^T Q(t) B = \begin{pmatrix} A_0 + tA_1 + t^2 A_2 & t(B_1 + tB_2) \\ t(B_1 + tB_2)^T & t(C_1 + tC_2) \end{pmatrix}. \tag{148}$$

Here $A_j = E^T Q_j E$, $B_j = E^T Q_j K$, $C_j = K^T Q_j K$; constant cross and kernel blocks are zero. Define

$$H(t) = \begin{pmatrix} A_0 + tA_1 + t^2 A_2 & \delta(B_1 + tB_2) \\ \delta(B_1 + tB_2)^T & \delta(C_1 + tC_2) \end{pmatrix} = H_0 + tH_1 + t^2 H_2. \tag{149}$$

Its three Bernstein controls in $t \in [0, \delta]$ are

$$B^{[0]} = H_0, \qquad B^{[1]} = H_0 + \delta H_1/2, \qquad B^{[2]} = H_0 + \delta H_1 + \delta^2 H_2. \tag{150}$$

The middle control is not the midpoint sample of H.

B.4 Exact positivity on the two parameter intervals

The file `theta_certificate/interval_witness.json` supplies six invertible lower-triangular rational preconditioners T_j, ordered as the three descent controls followed by the three coercivity controls. For each control form $G_j(u) = T_j B^{[j]}(u) T_j^T$, using the corresponding family for the index. The cover consists of exactly two closed cells:

$$\left[\frac{3341}{1000}, \frac{13367}{4000}\right], \qquad \left[\frac{13367}{4000}, \frac{1337}{400}\right]. \tag{151}$$

Each rational matrix entry is put over a common polynomial denominator; its numerator has degree at most fourteen.

Here is the complete interval rule used to prove each sign. On a cell $[a, b]$, expand a polynomial after $u = a + (b - a)s$ as $\sum_{j=0}^{d} a_j s^j$. Its Bernstein coefficients are

$$\beta_k = \sum_{j=0}^{k} a_j \frac{\binom{k}{j}}{\binom{d}{j}}, \qquad 0 \le k \le d. \tag{152}$$

Since the Bernstein basis is nonnegative and sums to one on $[0, 1]$, $\min_k \beta_k \le p(u) \le \max_k \beta_k$ on the entire cell. Constant polynomials use $d = 0$. Apply this rule separately to each numerator and denominator. Every denominator range stays strictly away from zero. Interval division, taking the minimum and maximum of the four endpoint quotients, then bounds each rational entry on the full cell. All operations are rational.

If $[\ell_{ij}, h_{ij}]$ encloses the (i, j) entry of a symmetric G_j, form its row bound

$$\gamma_i = \ell_{ii} - \sum_{j \ne i} \max\{|\ell_{ij}|, |h_{ij}|\}. \tag{153}$$

The exact data give $\gamma_i > 0$ for all 96 cases (six controls, two cells, eight rows). Gershgorin's theorem therefore proves that every $G_j(u)$, hence every unpreconditioned control, is positive definite on all of J. For an easily readable lower bound, the minimum computed rational row margin satisfies

$$\min \gamma_i > \frac{19}{50} > 0. \tag{154}$$

The full exact fraction is retained in the public witness and recomputed by both interval implementations; (154) is checked by exact cross multiplication. The proof never uses a rounded eigenvalue.

The same polynomial range rule proves nonnegativity of every multiplier at $t = 0$ and $t = \delta$. Of the 164 endpoint slots, 65 are strictly positive throughout J and the other 99 are identically zero. Affinity in t proves $\lambda, \mu \geq 0$ on $J \times [0, \delta]$. The original coefficient denominator checks and (147) ensure all expressions are defined throughout.

For each fixed u, Bernstein interpolation of (150) gives $H(t) > 0$ for $0 \leq t \leq \delta$. Its lower-right principal block shows $C_1 + tC_2 > 0$. With $S_t = \operatorname{diag}(I, (t/\delta)I)$ one has the exact identity

$$B^T Q(t) B = S_t H(t) S_t + \operatorname{diag}\big(0, t(1 - t/\delta)(C_1 + tC_2)\big). \tag{155}$$

Both implementations verify it coefficientwise through degree four. For $t > 0$ its first term is positive definite and the second is positive semidefinite. Thus $Q(t) > 0$. Since $r > 0$, this proves $D_8, C_8 > 0$, completing the positivity part of Lemma 2. All the remaining assertions were the exact identities and specialization checks already described. This proves the lemma on the full continuous rectangle.

B.5 Independent reconstruction and public source closure

The public suite is entered by `theta_certificate/verify_suite.py`. It requires Python 3.10 or later and SymPy; the tested SymPy version is 1.14.0. The old verifiers in the parent directory continue to use only Python's standard library. Both generations of checks are run by the root `verify_all.sh`. Assertions must remain enabled.

The author continuum checker uses direct binomial conversion to Bernstein coefficients. The independent interval checker reconstructs its own residual matrices and bases, performs affine substitution by exact Horner arithmetic, and obtains Bernstein coefficients by triangular inversion of the power-basis relation. The independent literal checker uses Hessians of scalar proximal-slack polynomials. None imports an author checker or accepts another program's saved matrices or pass status. Independent endpoint and period-three programs additionally reconstruct the normal tests; the general-coupling check retains a symbolic rectangular spatial operator and verifies both cyclic orientations. The Hilbert-space implications remain the analytic arguments in the body, not a conclusion from a finite-dimensional numerical experiment.

Only the three raw files `family.json`, `eighth_certificate.json` and `interval_witness.json` in the supplement supply mathematical inputs. Their schema, exact provenance and payload hashes are documented there. Descriptive saved-status metadata is not proof input. No optimization solver, pickle, discovery state, private source directory, cached matrix or network access is required. The source allowlist and noncyclic manifests identify every public file. The historical root `SHA256.json` retains its original three-file

responsibility; the new top-level manifest covers the whole public second-version payload except itself. Hash agreement establishes identity, not mathematical validity, so exact replay is still required.

The old fixed-parameter coefficient data and theorem/witness notes are retained byte for byte. New notes have distinct general-theta names. The local current-source record identifies all release inputs without requiring an earlier report bundle. The source archive's internal PDF keeps its stable filename and has the same bytes as the externally named second-version PDF.

C Exact replay of the additional regions and five cycle

The supplementary inputs use the same ordered slots, raw convexity slacks, multiplier support, kernel complements and desingularization identities specified in Appendix B. The replay contains no numerical optimization, saved solver state or sampled positivity inference. All comparisons below are exact rational or algebraic comparisons.

C.1 Continuous positivity and section limitations

The directory `theta_widening` contains a byte-identical copy of `theta_certificate/family.json`, the single-cell interval witness and two exact descent obstructions. For each of the six controls in (150), its rational congruence matrix is listed in the interval witness. Applying the polynomial Bernstein range rule (152) and exact rational division on J_{w} gives a minimum Gershgorin margin

$$m_{\mathrm{w}} = 0.01458749735\ldots > \frac{1}{100}. \tag{156}$$

The JSON stores the full rational value; the decimal is only explanatory. Every denominator has a certified sign. The same calculation checks all 164 multiplier endpoints (82 multipliers at $t = 0, \delta$), including identically zero endpoints, the kernel-basis determinants, positivity of $R - \delta$ and $\theta'(u)$, and both metric diagonal identities. The degree-four congruences are checked coefficientwise. Independent scalar-Hessian reconstruction and an independently implemented Bernstein conversion reproduce these claims without importing the author verifier. The 288 original coefficient slots, including zero multipliers, are recovered exactly at u_0.

For comparison, `OBSTRUCTIONS.json` gives rational vectors supported on the first four history slots at

$$u = \frac{16691}{5000} \quad \text{and} \quad u = \frac{33433}{10000}, \qquad t = \frac{1}{10^7}. \tag{157}$$

At each point the exact forms against both D_0 and $rD = D_0 + tD_1 + t^2 D_2$ are negative. The latter is an actual failure of this section's descent matrix at a positive subcritical

displacement, not merely a negative sufficient Bernstein control. Both points map inside $\theta \in [1/9, 1/7]$, but outside $\mathcal{I}_{\mathrm{w}}$. The independent verifier evaluates the scalar storage difference minus the fifty literal slack terms directly. These observations do not identify a maximal certified interval, rule out other free coordinates, or imply nonconvergence of the iteration.

C.2 Continuous replay of the different section

The only exact coefficient and positivity inputs for the different section are `theta_shifted/linear/family.json` and `INTERVAL_PROOF.json` in the same directory. The 86 free coordinates are rational affine functions selected between two rational anchors; all dependent coordinates are solved exactly. The resulting 189 supplied rational coefficient expressions have no poles on J_{s}. This construction does not use the old-coefficient regression condition.

The author and independent scalar-Hessian reconstructions check the literal matrix and value-flow identities, the common endpoint kernels, the metric diagonal identities and both kernel-complement determinants. Six rational congruences and independent exact Bernstein bounds on the single full cell J_{s} give a weakest Gershgorin lower bound

$$m_{\mathrm{s}} = 0.6455520725\ldots > \frac{3}{5}. \tag{158}$$

The full rational margin is stored in the witness. Of the 164 multiplier endpoints, 65 are strictly positive and 99 are identically zero; affineness in t gives nonnegativity throughout $[0, \delta]$. The same coefficientwise degree-four congruence as (155) then proves $D_8, C_8 > 0$ for every positive $t \le \delta$. Exact endpoint normal tests verify the matching triangle construction across the whole interval. No rejected section or numerical search output is a trusted input to these checks.

For Corollary 10.1, the separate input `theta_shifted/family.json` is specialized at $u = 17/5$. The public independent literal reconstruction is followed by `verification/verify_anchor.py` in that directory. It checks all coefficient denominators, the exact desingularization identities, six lists of eight positive rational LDL^T pivots, all 164 multiplier endpoints, endpoint ranks four and six, and the constant-history metric identities. The controls certify all $t \in (0, \delta]$, and contraction embedding gives every smaller actual product. The endpoint replay supplies the sharpness witness. This fixed-anchor proof uses no continuous extrapolation claim.

C.3 Algebraic signs for the five cycle

The directory `theta_five_cycle` contains two complementary replays. The first reconstructs (102) from (105), checks the physical rotation residuals, and bounds every normal test by rational enclosures for $\sqrt{5}$, $\sqrt{10 + 2\sqrt{5}}$ and $\sqrt{10 - 2\sqrt{5}}$. It obtains

$P_j, H_j < -1/10000$ at the center and at the relevant parameter endpoints, so the affine interpolation used in Theorem 7 is exact.

The independent implementation uses only the Python standard library. Set $a = \sqrt{10 + 2\sqrt{5}}$ and reduce field elements modulo $a^4 - 20a^2 + 80 = 0$. The enclosure

$$\frac{3804226065180614}{10^{15}} < a < \frac{3804226065180615}{10^{15}} \tag{159}$$

is certified by opposite polynomial signs and the positive derivative $4a(a^2 - 10)$ above $\sqrt{10}$. The rotation entries are

$$\cos\phi_1 = (a^2 - 12)/8, \quad \sin\phi_1 = a/4, \qquad \cos\phi_2 = (8 - a^2)/8, \quad \sin\phi_2 = a(a^2 - 12)/16.$$

The diagonal change of coordinates $T = \mathrm{diag}(\sqrt{A_1}, \sqrt{A_1}, \sqrt{A_2}, \sqrt{A_2})$ removes coordinate square roots and gives the rational physical metric $G = T^T T$. Because T commutes with the rotations, the transformed states $(1, 0, 1, 0)$ and $(c_1/A_1, d_1/A_1, c_2/A_2, d_2/A_2)$ represent exactly (106).

Starting with these physical states, the independent checker verifies isometry, fifth order, all forty central normal inequalities and all 160 corner inequalities separately. It also checks positive pairwise primal distances, zero state sums and positive four-by-four edge Gram determinants for both hulls. These direct checks do not assume the moment characterization. They prove the literal map, including the wraparound transition, and verify the threshold comparisons with rational arithmetic.

C.4 Reproduction and preservation

Run `bash verify_all.sh` from the extracted source directory, with Python assertions enabled and SymPy as specified in `requirements.txt`. This runs the two original fixed-parameter checks, all eight original general-parameter checks, and the additional author and independent replays. The wrapper uses a temporary copy for generated outputs, so verification cannot alter the frozen public payload. `check_hashes.py` checks the public allowlist manifest; `verify_preservation.py` checks the original mathematical body and every historical label. The new sources neither renumber the original Theorem 1 nor change the meaning of (2).

The build command is `bash build.sh`. It requires pdfLaTeX and the packages named in the unchanged style file, uses `-no-shell-escape`, and includes a system font-map fallback covering `cm`, `cmextra`, `latxfont`, `symbols` and `euler`. No optimization or network service is needed to reconstruct the proof or typeset the article.

D Exact replay of the five cycle phase

The public directory `theta_phase/verification/` contains only release-local exact proof code and ancillary coefficients. Its driver reconstructs the polynomial, checks the base root, verifies the literal geometry, certifies the entire continuation and crossing calculation, and then certifies endpoint rigidity. Python's standard library and SymPy suffice; no numerical optimization package or solver is involved. The main release replay retains every earlier check and runs these additional checks in a temporary copy of the public whitelist. Cached success fields are not proof inputs, and the shipped files are hash-checked before and after the replay.

The polynomial file lists the rational normalization of (119). Rebuilding the cubic pencil proves exact proportionality to that discriminant and recovers its bidegree and 195-term count. At $\theta = 9/50$, the primitive polynomial in r has degree twelve. Exact Sturm isolation selects its root in $(\ell, \ell + 10^{-60})$, where

$$\ell = 2 + \frac{9390118422062277595500536872126}{10^{30}} + \frac{4246537518324504006665389443 91}{10^{60}}.$$

The stored coefficients and this isolator are regenerated. Exact polynomial reductions verify the repeated-root identities, and rational outward intervals verify every required sign. The very narrow interval is convenient for algebraic evaluation; the decimal approximation in (112) carries no logical role.

For the continuum, the 1,024 rectangles in (120) cover all of I_5. Every horizontal root-boundary sign in (121) is checked uniformly over its entire θ interval, using a centered value and an interval bound on $\mathcal{F}_\theta$ times the displacement from the midpoint. Every derivative sign $\mathcal{F}_r > 0$ is checked on the full rectangle. The remaining rational witness and dual expressions are enclosed using interval automatic differentiation: a scalar expression carries an enclosed center value and enclosures of both first partial derivatives on the whole box. The multivariate mean-value formula bounds its complete range; product and quotient rules propagate the derivatives. Every divisor is proved bounded away from zero. This is a first-order enclosure with rigorous derivative bounds, not a truncated approximation with an omitted remainder.

Each rational interval operation rounds its lower endpoint down and its upper endpoint up on a fixed decimal rational grid, by integer division, including for negative values. Multiplication takes all four endpoint products. Reciprocation requires exclusion of zero. The sign decisions use only these rational bounds. The replay checks strict positivity of X_k, B_k, z, the denominator and rank conditions, and every active dual weight; it checks strict negativity of all three inactive rows. In the $u_3 = -1$ convention its bounds in particular give $\lambda_i > 1/1000$ and $P_2, P_4, H_4 < -17/100$ throughout the cover. These are certified lower margins, not claimed optimal margins.

A separate rigidity pass checks exact adjacency, overlap, base-root inclusion and the ordering (132). It proves (130) on 128 rational boxes covering the graph. Subdivision is

used only to tighten inconclusive enclosures; a failed interval test is never interpreted as a proof of infeasibility. The graph-gluing argument in Section 11.2 specifies why these local boxes define a single analytic branch.

The two comparison polynomials in (131) are regenerated by exact substitution and elimination. Sturm sequences count one root of each in the entire I_5, and isolate the roots in disjoint rational intervals. An independently reconstructed check verifies squarefreeness, the full cover, normalized ray rescaling, and all literal residual identities. In particular, direct real rotation multiplication retains an arbitrary Gram-slack coordinate and checks the $A = 0$ convention separately. Rotation equivariance yields all forty off-diagonal vertex tests and exact zero centroids.

The accompanying source mapping records the original proof-source hashes and any mechanical release-path adaptations. Internal discovery and review directories are not dependencies or public inputs. The finite exact checks certify the displayed mathematical identities and inequalities; the prose arguments above supply the passage to arbitrary Hilbert spaces, global exclusion below the branch, fixed convex realizations and the precise finite-period scope.

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[9] Report 24, *An exact two-mode characterization of Chambolle–Pock five-cycles*, earlier research manuscript.

[10] Report 25, *An exact finite-period Fourier–SOC criterion for primal-first Chambolle–Pock cycles*, earlier research manuscript.

[11] Report 27, *Exact attained five-cycle thresholds for primal-first Chambolle–Pock dynamics*, earlier research manuscript.